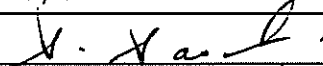
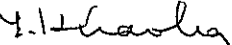
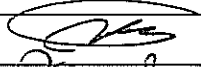
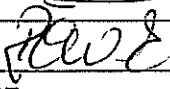


DUBAI RAPID LINK CONSORTIUM

DUBAI METRO PROJECT OFFICE

Contract No.: DM001		CDRL No.:		
Project Title: DUBAI METRO		CDRL Title:		
Document Title: Jebel Ali Auxiliary Depot Structure Design Calculation Building 21 – Inflammable Material Storage DCP 3 Submission				
Revision History				
A3	15-May-07	Second Issue	Signed below	Signed below
A2	14-Nov-06	First Issue	Signed below	Signed below
MARK	DATE	DESCRIPTION	RAIL	CIVIL
APPROVED		Project Director	S. Shimamura N/A	
		Deputy Project Director	S. Sasaki 	
		Checked By (Project Director)	Y. Hiraoka 	
		Checked By (QA/QC Manager)	J. Whorlow 	
Checked By	N/A	Checked By (Project Manager)	N/A	
Checked By	N/A	Checked By (Engineering Manager)	M. Makino 	
Prepared By	N/A	Issued By	B. Maney 	
DATE	-	DATE	15 May 2007	
RAIL SYSTEM			CIVIL JV	
CONTRACTOR'S DOCUMENT No.:		DOCUMENT No.:		REVISION
		DM001-E-AAK-CDP-DR-DCC-373906		A3

**Dubai Metro, UAE
Jebel Ali Depot**

**Structural Design Calculation
(Enhanced Design)
Building 21 – Inflammable Material Storage**

May 2007

Client: JT Metro JV	Contract No. (if any):
Project Title: Dubai Metro, UAE	Project No.: JRD4561
Document No.: DM001-E-AAK-CDP-DR-DCC-373906	Controlled Copy No.:
Document Title: Auxiliary Depot- Jebel Ali Structural Design Calculation (Enhanced Design) Building 21 – Inflammable Material Storage	
Covering Letter / Transmittal Ref. No.: To be advised	Date of Issue: 02 May 2007

Revision, Review and Approval/Authorization Records

A3	Issued for Approval (Enhanced Design)	CK Lam May 07	P. Shepherd-Smith May 07	B. Maney May 07
A2	Issued for Approval	CK Lam Oct 06	P. Shepherd-Smith Oct 06	B. Maney Oct 06
A1	Internal Review	CK Lam Sep 06	P. Shepherd-Smith Sep 06	B. Maney Sep 06
Revision	Description	Prepared by / date	Reviewed by / date	App. or Auth. by / date

Distribution (if insufficient space, please use separate paper)

Controlled Copy No.	Issued to
1	JT Metro JV
2	Atkins (Office Copy)
3	Atkins (P. Shepherd-Smith)
4	Meinhardt Infrastructure (CK Lam)

Note: App. and Auth. mean "Approved" and "Authorized" respectively.

Document No.: DM001-E-AAK-CDP-DR-DCC-373906

Rev A3

Date: 02 May 07

Page

Table of Contents

1	TYPICAL PORTAL FRAME TRANSVERSE ANALYSIS.....	001
2	FOUNDATION DESIGN CHECK	040
3	CONNECTIONS DESIGN	045

1 TYPICAL PORTAL FRAME TRANSVERSE ANALYSIS

Document No.: **DM001-E-AAK-CDP-DR-DCC-373906**

Rev A3

Date: 02 May 07

Page



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Job No

Sheet No

1

Rev

Part

Job Title

Ref

By

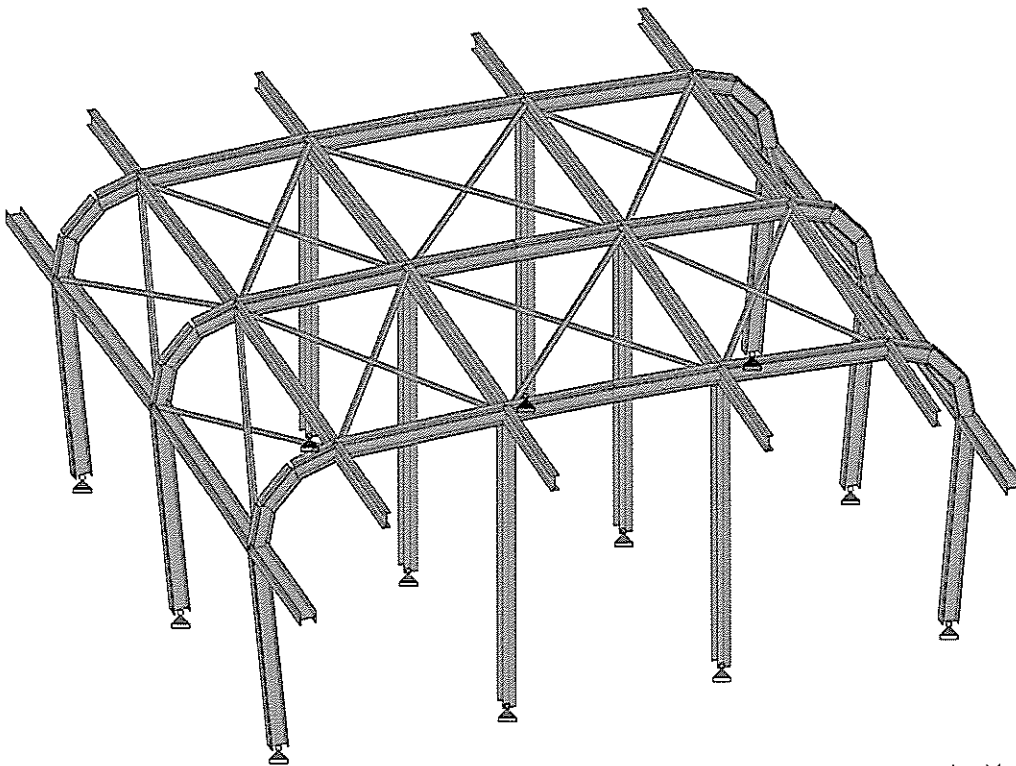
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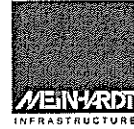
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Date/Time 02-May-2007 15:02



INFLAMMABLE MATERIAL STORE (JEBEL ALI)



Site:	Dubai Metro Project	Date:	28-Apr-07	Job No.:	68010.01
	Inflammable Material Store				
	Loadings	Designed by:	Rudy	Sheet No.:	

Roof Loadings :

DL

 W_{DL} :

Sheeting	0.1 kPa		
Purlins	0.1 kPa		
Insulation	0.3 kPa		
M&E Finishes	0.3 kPa		
	<u>0.8 kPa</u>	x 1 =	0.8 kN/m

LL

0.6 kPa x 1 = 0.6 kN/m

P_{LL} : A/C equipment S/W
 = 5 kN

(point load at mid-span of bay 2, 4)

Notional horizontal load, Ph

(Load Case of 1.4DL + 1.6LL)

Tributary load area for the roof

= $(10+2*3)*(12+2*1.5)$ = 240 m²

Total Dead Load

192 kN

Total Live Load

149 kN

(Including A/C equipment)

Tributary load area for the wall

= $6*2*(10+2*3+12+2*1.5)$ = 372 m²

Total Dead Load

186 kN

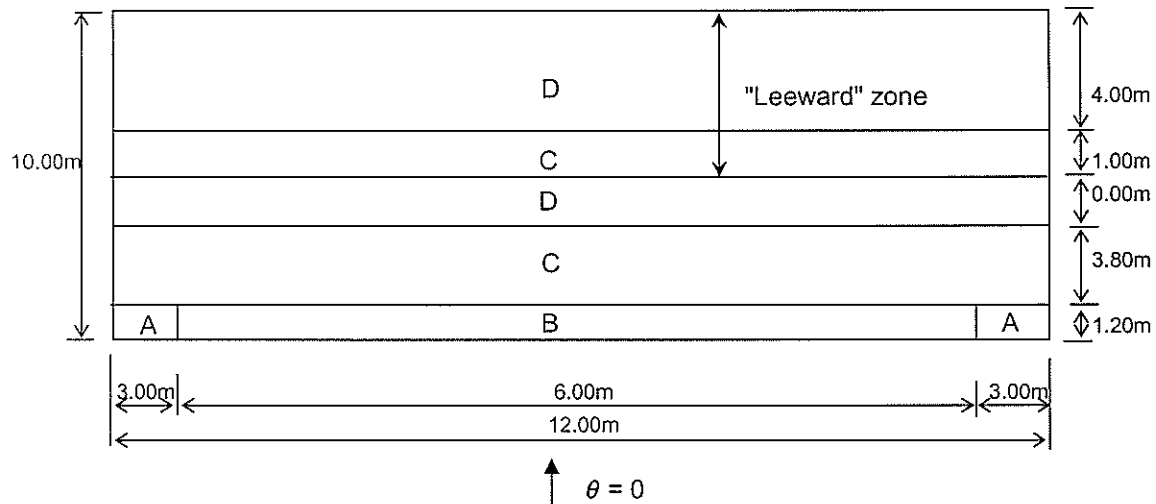
Total Live Load

(excluding the M&E Finishes)



Site:	Dubai Metro Project	Date:	28-Apr-07	Job No.:	68010.01
	Imflammable Material Store				
	Wind Load Computation	Designed by:	Rudy	Sheet No.:	

Average Cpe for roof (Wind direction normal to eave)

PLAN

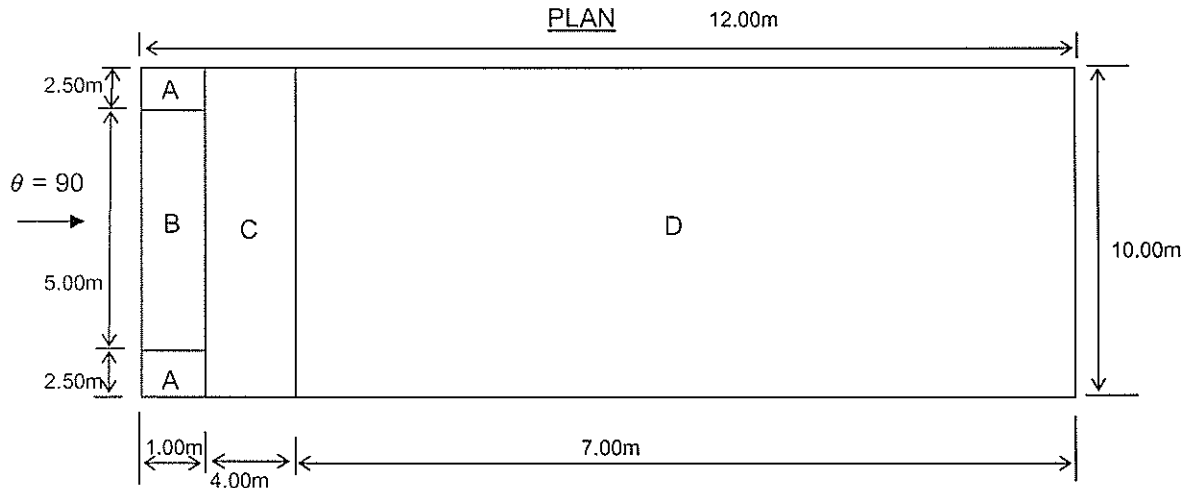
Design as flat roof (with sharp eaves)

$$\begin{aligned}
 B &= 12 \text{ m} \\
 H &= 6 \text{ m} \\
 b &= \text{smaller of } (B, 2H) \\
 &= 1 \times 12 \\
 &= 12 \text{ m}
 \end{aligned}$$

Zone	Windward		Leeward		
	Area (m ²)	Cpe	Area	Cpe1	Cpe2
A	7.2	-2			
B	7.2	-1.4			
C	45.6	-0.7	12.0	-0.7	-0.7
D	0.0	-0.2	48.0	-0.2	0.2
Total	60.0		60.0		
Average		-0.94		-0.30	0.02



Site:	Dubai Metro Project	Date:	28-Apr-07	Job No.:	68010.01
	Inflammable Material Store				
	Wind Load Computation	Designed by:	Rudy	Sheet No.:	

Average Cpe for roof (Wind direction normal to gable end)


Design as flat roof (with sharp eaves)

$$\begin{aligned}
 B &= 10 \text{ m} \\
 H &= 6 \text{ m} \\
 b &= \text{smaller of } (B, 2H) \\
 &= 1 \times 10 \\
 &= 10 \text{ m}
 \end{aligned}$$

For maximum uplift, assume zone C coefficient =>

$$C_{pe} = -0.7$$

For maximum downward force, assume zone D coefficient =>

$$C_{pe} = 0.2$$

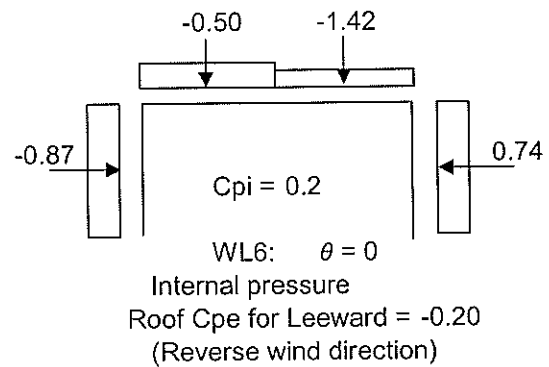
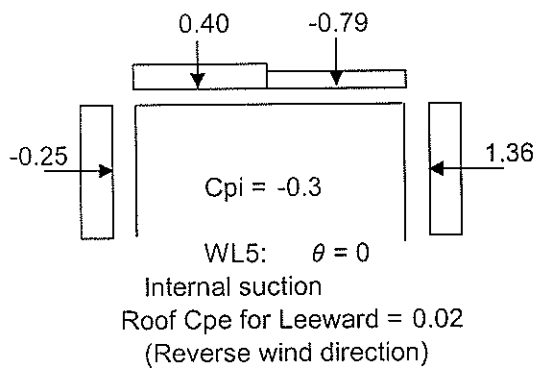
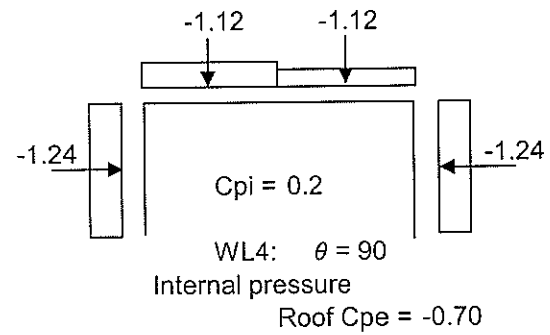
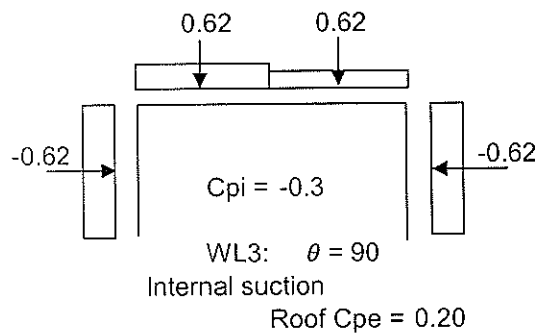
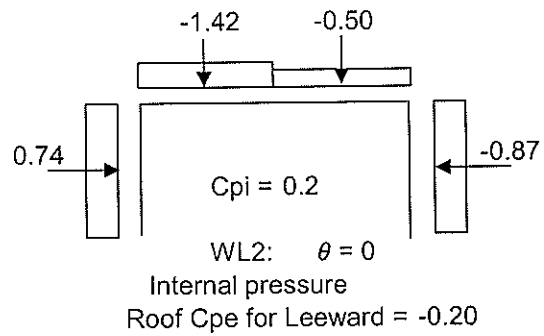
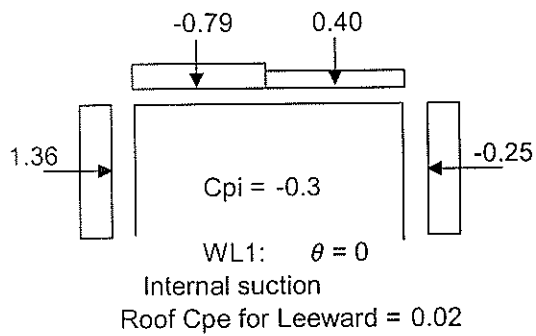
Wind Load Computation

$$\begin{aligned}
 D &= 10.00 \text{ m} \\
 H &= 6 \text{ m} \\
 D/H &= 1.67
 \end{aligned}$$

Elements	Effective Wind Speed V_e m/s	Dynamic Wind Pressure q kN/m ²	External Pressure Coefficient, C_{pe}			
			$\theta = 0$ (normal to eave)		$\theta = 90$ (normal to gable end)	
			Windward	Leeward		
Roof	45	1.24	-0.94	-0.30 0.02	-0.7 (Zone C)	0.2 (Zone D)
Wall	45	1.24	0.79	-0.5 -0.5	-0.8 (Zone B)	-0.8 (Zone B)

Site:	Dubai Metro Project Imflammable Material Store Wind Load Computation	Date:	28-Apr-07	Job No.:	68010.01
		Designed by:	Rudy	Sheet No.:	

$$\text{Wind Load} = 1 \times q \times (C_{pe} - C_{pi}) \quad \text{kN/m} \quad (\text{For unit width})$$



Site: Dubai Metro Project
Inflammable Material Store
Earthquake Loading

Date: 28-Apr-07
Designed by: OEH

Job No.: 68010.01
Sheet No.:

Estimation of Earthquake Loading

Total Dead Load on the Building Column W = Given below (including part of Live Load)
Height of the Column h_n = Given below

Design Information:

- 1) Seismic Zone = 2A (from Design Brief)
2) Occupancy Category = 1 (Tanle- 16-K, Conservatively adopt 1)
3) Lateral Force procedure = **Static Procedure** (Section 1629.8 seismic zone 2A, structures not more than 5 stories)

Earthquake Loads

- 4) **Base Shear** $V_{base-S} = (C_v \times I \times W) / (R \times T)$ (Section 1630.2.1)
Seismic Coefficient $C_v = 0.32$ (Table 16-R assuming soil profile type S_D)
Importance Factor $I = 1.00$ (Table 16-K)
Numeric. Coef: Strength & Ductility $R = 4.50$ (Table 16-N for OMRT steel only)
Elastic Fundamental Period of Vibration $T = C_t (h_n)^{3/4}$ Sec.
Numerical Coefficient $C_t = 0.0853$ (Steel frame t.b.c)

- 5) **Limiting Values of Base shear** $V_{base-S-Max} = (2.5 \times C_a \times I \times W) / R = 55.31$
 $V_{base-S-Min} = 0.11 \times C_a \times I \times W = 10.95$

Design Earthquake Loads

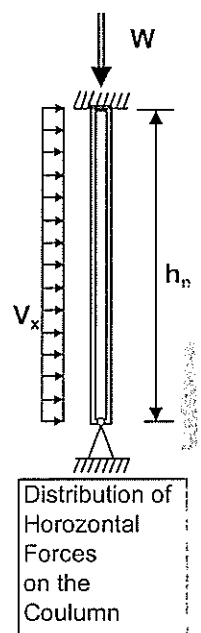
- (Section 1630.6)
 $E_x = \rho \times \text{Min. } (E_h \text{ or } V_{base-S-Max}) + E_v$
 $E_{x-max} = \Omega \times \text{Min. } (E_h \text{ or } V_{base-S-Max})$
 $V_{base-S} = E_h =$ Earthquake load in the element due to the base shear, V , as set out above (for non-structural components refer to section 1632)
Vert. component of EQ ground motion $E_v = 0.5 C_a I D$ (Table 16-K)
Seismic Coefficient $C_a = 0.22$ (Table 16-Q, for soil type S_D)
Seismic Force Amplification Factor $\Omega = 2.80$ (Table 16-N)
Dead Load on the element $W = D =$ Given below
 $\rho = 2 - 6.1 / (r_{max} \sqrt{A_B}) = 1.25$ (conservative) ($1.0 \leq \rho \leq 1.5$)
 $r_{max} =$ The maximum element-storey shear ratio
 $A_B =$ Ground floor area of the structure

Distribution of Horizontal Forces:

Distribution of Horizontal Force on the Column $V_x = E_x / h_n$

Col	Gr. Load, W , kN	Ht h_n , m	Time, T sec.	V_{base-S} kN	V_{base-S} kN	E_v kN	E_x kN	E_{x-max} kN	V_x kN/m	V_{x-max} kN/m
C1	453	6.00	0.33	98.4	55.31	49.775	118.9	154.9	19.8	25.8

Load per column = $6 \times 19.8 / 8$
= 15 kN





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Job No

Sheet No

1

Rev

Part

Job Title

Ref

By

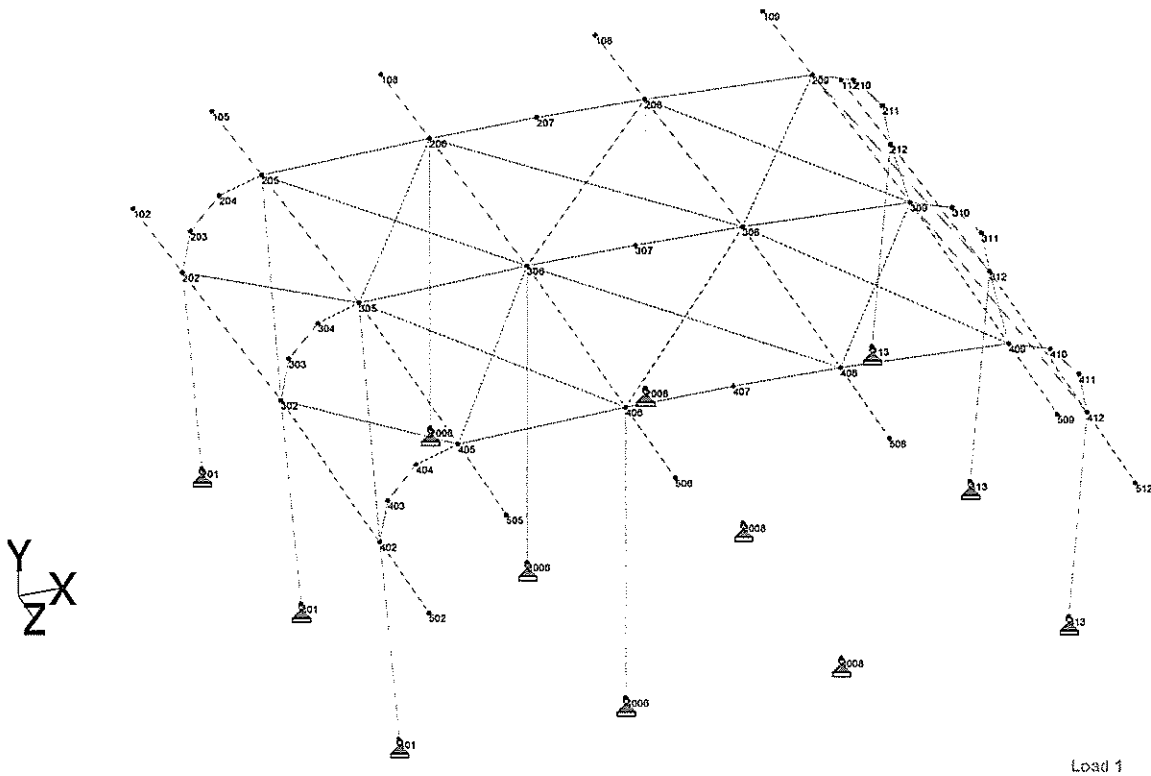
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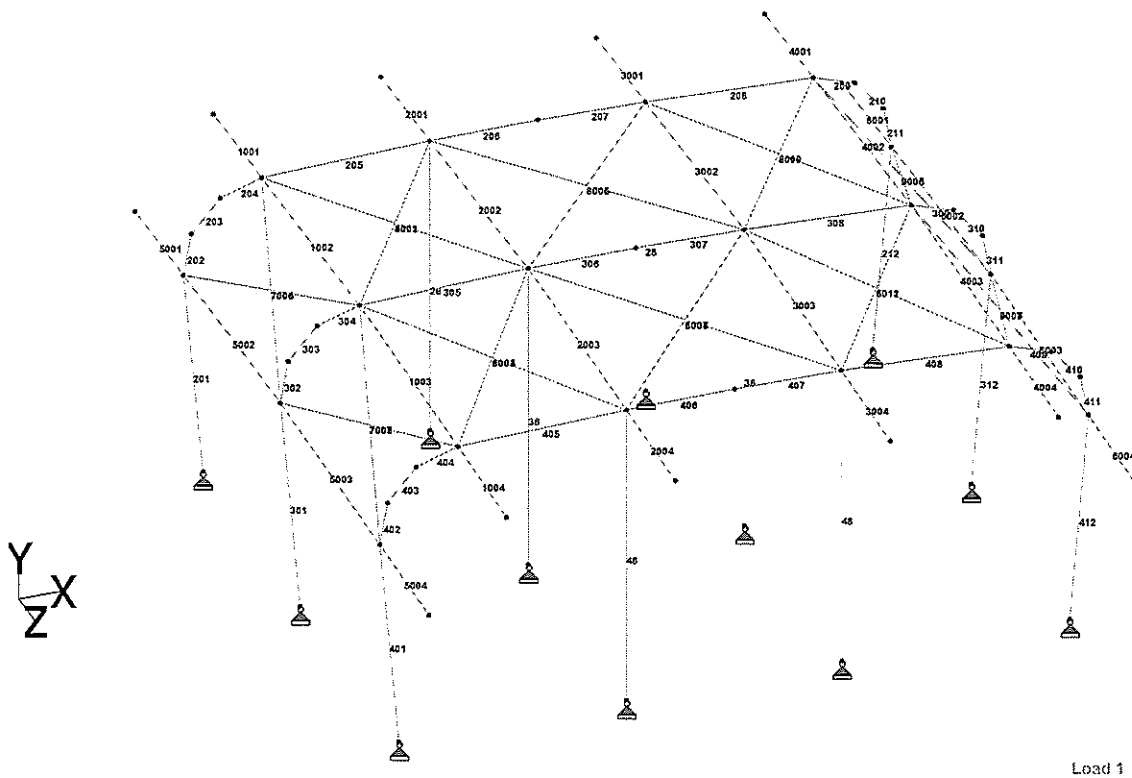
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Job No	Sheet No 1	Rev
Part		
Ref		
By	Date April 2007	Chd
Client	File Inflam-Mat_Store.std	Date/Time 02-May-2007 15:02





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Job No

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1

Rev

Part

Job Title

Ref

By

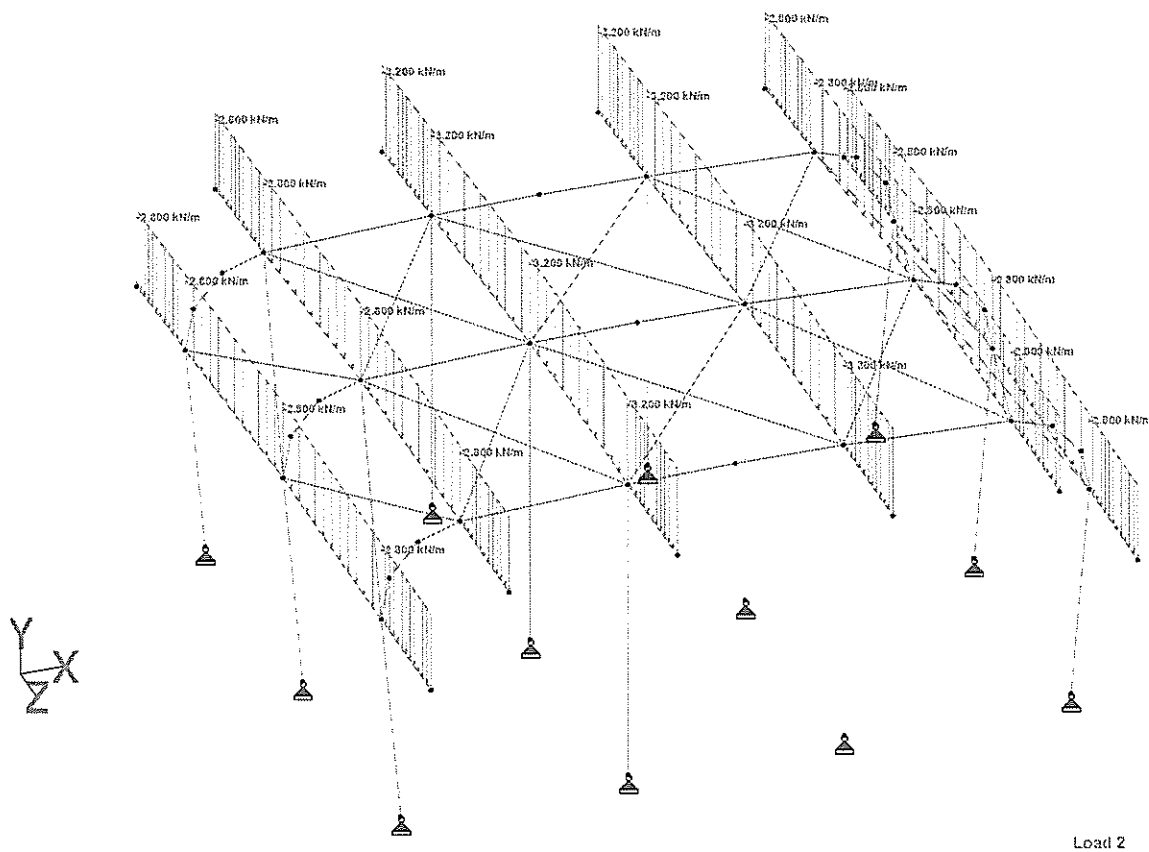
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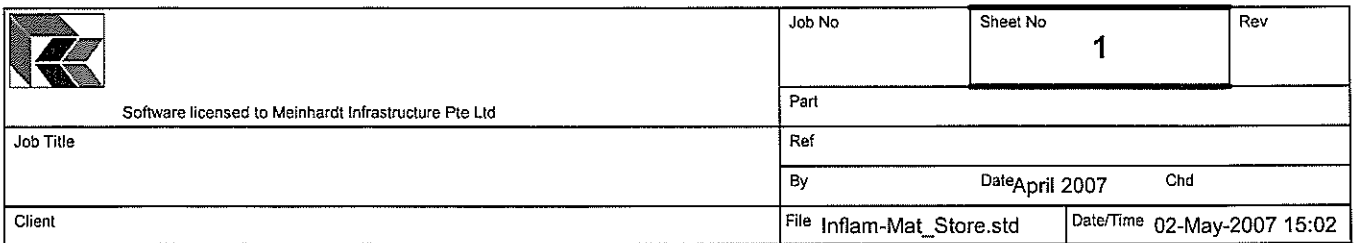
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File Inflam-Mat_Store.std

Date/Time 02-May-2007 15:02





	Job No	Sheet No 1	Rev
	Part		
Software licensed to Meinhardt Infrastructure Pte Ltd	Ref		
Job Title	By _____ Date April 2007 Chd _____		
Client	File Inflam-Mat_Store.std	Date/Time 02-May-2007 15:02	

	Job No	Sheet No 1	Rev
	Part		
Software licensed to Meinhardt Infrastructure Pte Ltd	Ref		
Job Title	By _____ Date April 2007 Chd		
Client	File Inflam-Mat_Store.std	Date/Time 02-May-2007 15:02	

	Job No	Sheet No 1	Rev
	Part		
Software licensed to Meinhardt Infrastructure Pte Ltd	Ref		
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Client	File Inflam-Mat_Store.std	Date/Time 02-May-2007 15:02	

	Job No	Sheet No 1	Rev
	Part		
Software licensed to Meinhardt Infrastructure Pte Ltd	Ref		
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	Part		
Software licensed to Meinhardt Infrastructure Pte Ltd	Ref		
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	Part		
Software licensed to Meinhardt Infrastructure Pte Ltd	Ref		
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	Part		
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	Part		
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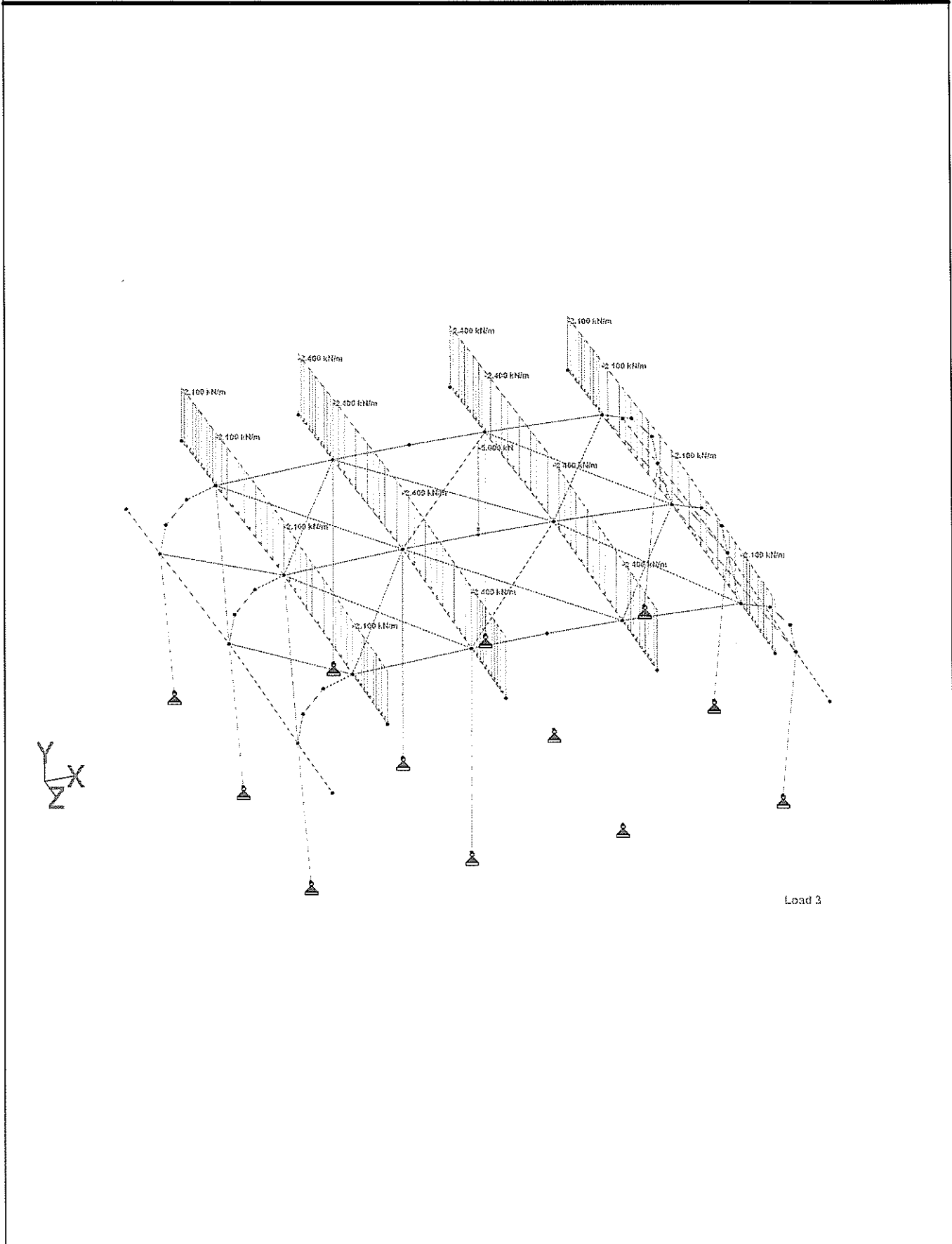
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	Part		
Software licensed to Meinhardt Infrastructure Pte Ltd	Ref		
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Client	File Inflam-Mat_Store.std	Date/Time 02-May-2007 15:02	

	Job No	Sheet No 1	Rev
	Part		
Software licensed to Meinhardt Infrastructure Pte Ltd	Ref		
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Client	File Inflam-Mat_Store.std	Date/Time 02-May-2007 15:02	

	Job No	Sheet No 1	Rev
	Part		
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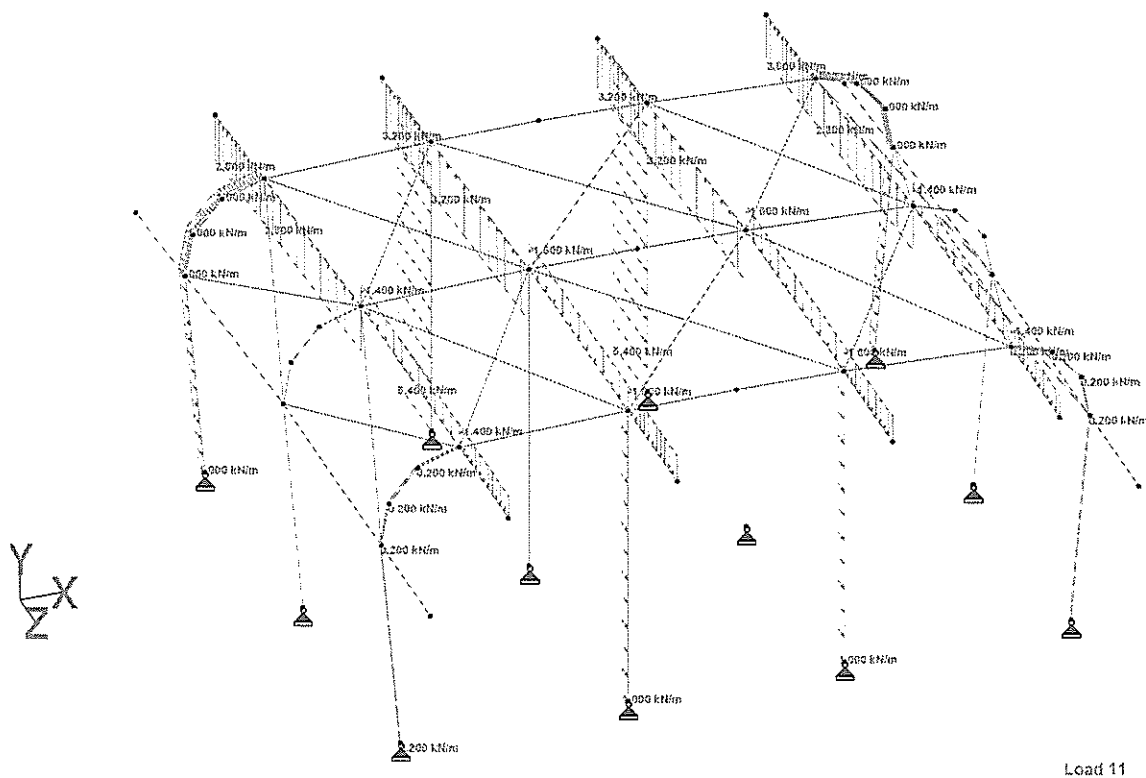
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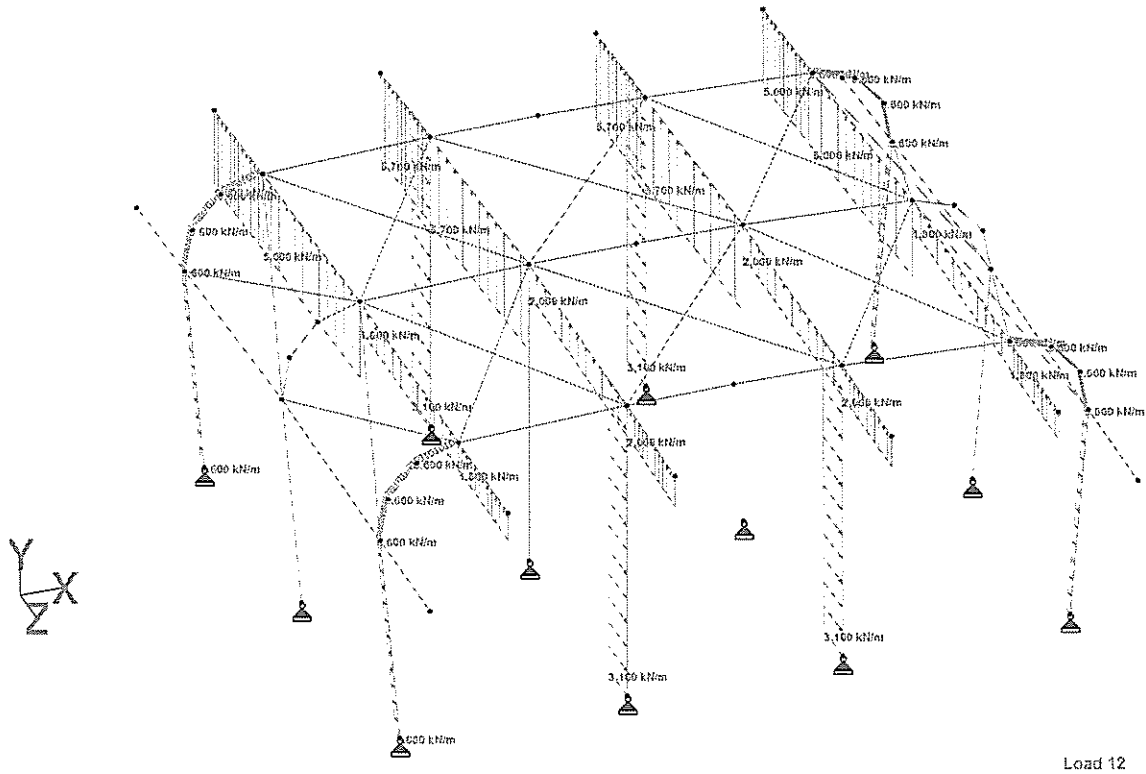
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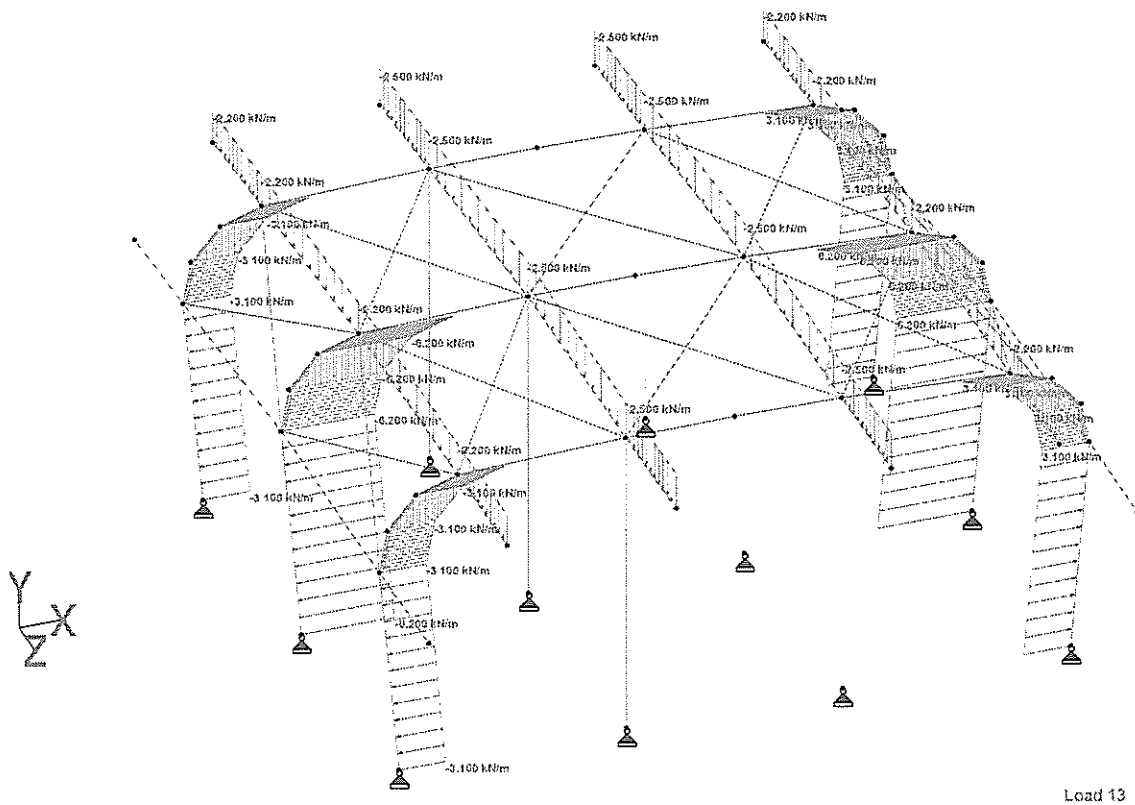
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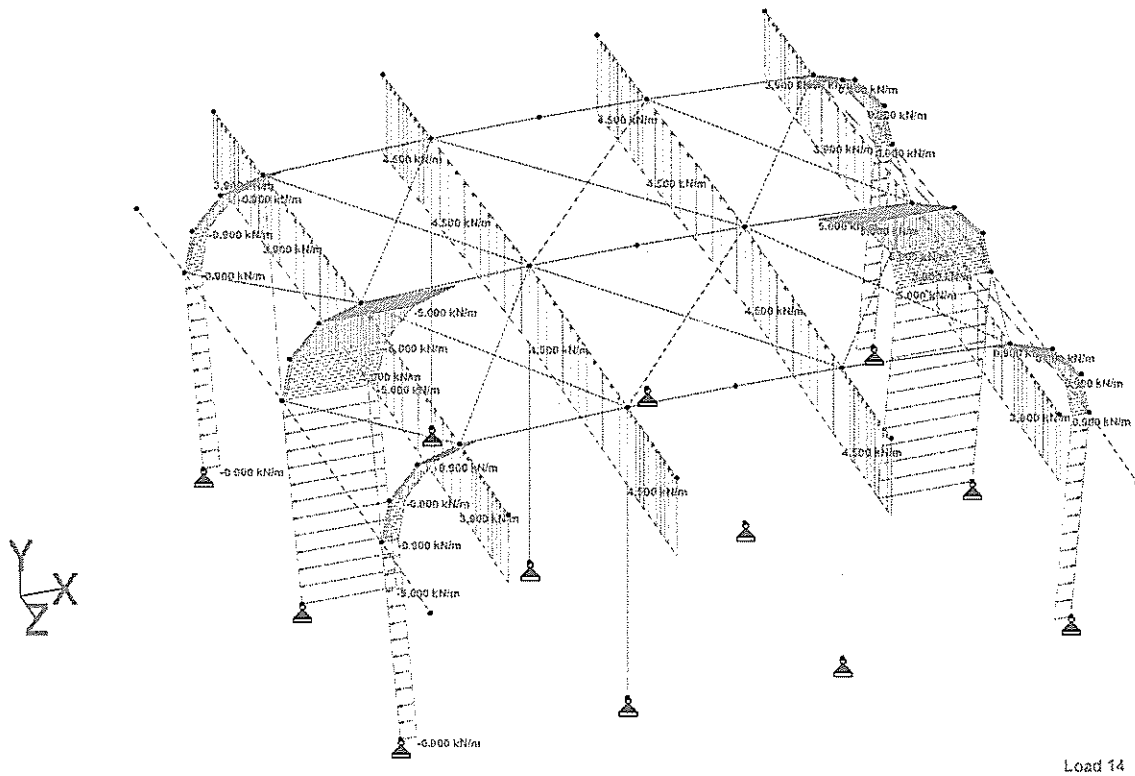
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Part		
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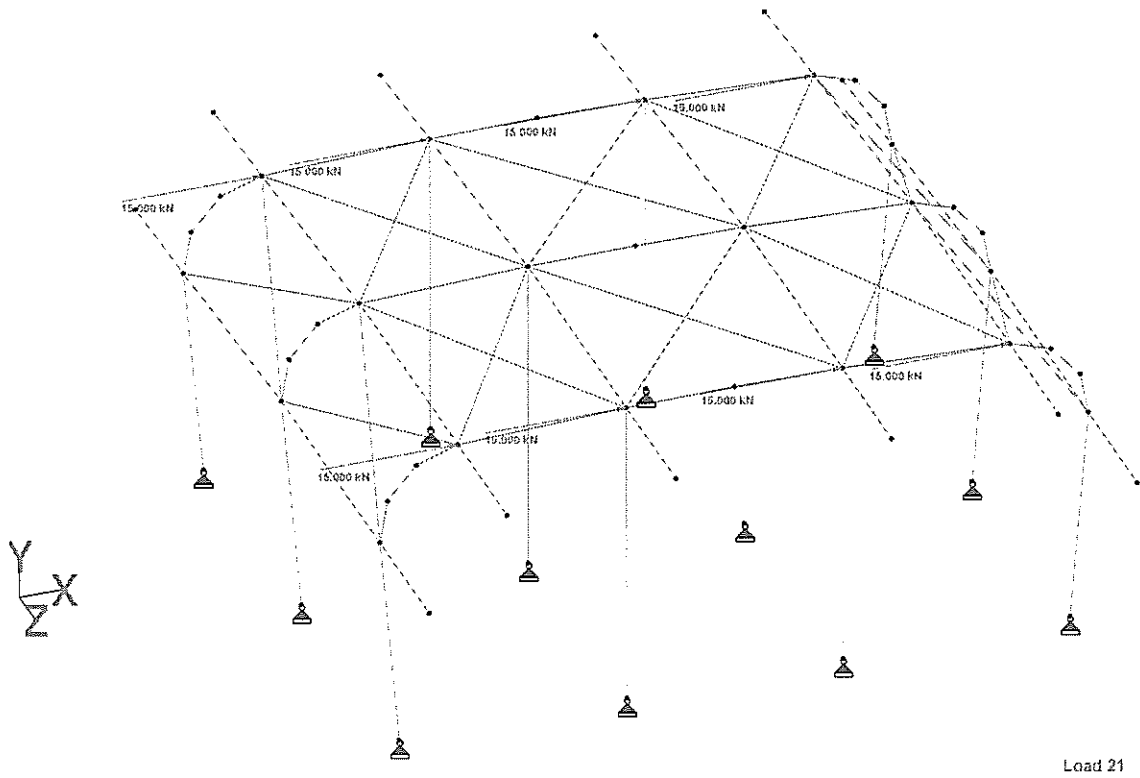
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Part		
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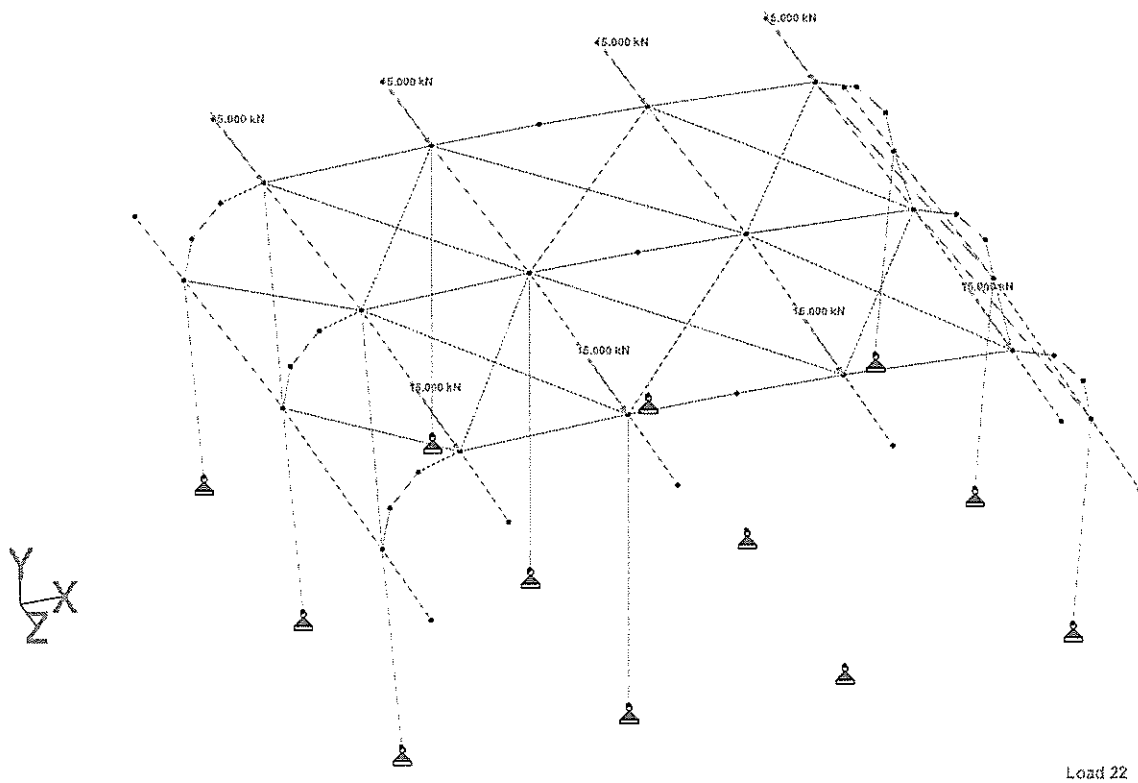
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Part		
Ref		
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Client	File Inflam-Mat_Store.std	Date/Time 02-May-2007 15:02





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Job No	Sheet No 1	Rev
Part		
Ref		
By	Date April 2007	Chd
Client	File Inflam-Mat_Store.std	Date/Time 02-May-2007 15:02



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4. END JOB INFORMATION
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44.
45. 26 2006 206; 28 2008 208
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61. 8005 206 306; 8006 306 206; 8007 306 406; 8008 406 306
62. 9009 208 309; 9010 308 209; 9011 308 409; 9012 408 309
63. 9005 212 309; 9006 312 209; 9007 312 409; 9008 412 309
64. DEFORM MATERIAL START
65. DEFORM MATERIAL
66. F 2.05E+008
67. F 2.05E+008
68. F 2.05E+008
69. POISSON 0.3
70. DENSITY 76.8195
71. ALPHA 1.2E-005
```

```
72. DAMP 0.03
73. END DEFINE MATERIAL
74. MEMBER PROPERTY BRITISH
75. 26 28 36 46 48 TAPERED 0.3 0.01 0.3 0.25 0.015 0.25 0.015
76. 201 TO 212 301 TO 312 -
77. 401 TO 412 TAPERED 0.3 0.01 0.3 0.25 0.015 0.25 0.015
78. 1001 TO 1004 2001 3001 3004 -
79. 4001 TO 4004 TAPERED 0.3 0.006 0.3 0.15 0.008 0.15 0.008
80. 2002 3002 3003 TAPERED 0.3 0.006 0.3 0.2 0.010 0.2 0.010
81. *
82. 5001 TO 5004 TAPERED 0.3 0.01 0.3 0.25 0.015 0.25 0.015
83. 6001 TO 6004 TAPERED 0.3 0.01 0.3 0.25 0.015 0.25 0.015
84. 7005 TO 7008 8001 TO 8012 9005 TO 9008 TABLE ST UA90X90X6
85. *7001 TO 7008 9001 TO 9008 TABLE ST UA90X90X6
86. *CONSTANTS
87. *
88. BETA 90 MEMB 26 28 36 46 48
89. *
90. BETA 90 MEMB 5001 TO 5004 6001 TO 6004
91. MATERIAL STEEL ALL
92. MEMBER TENSION
93. *7001 TO 7008 8001 TO 8012 9001 TO 9008
94. 7005 TO 7008 8001 TO 8012 9005 TO 9008
95. SUPPORTS
96. 201 213 301 313 401 413 2006 2008 3006 3008 4006 4008 PINNED
97. *
98. LOAD 1 SELFWEIGHT
99. SELFWEIGHT Y -1
100. LOAD 2 SDL
101. MEMBER LOAD
102. 1001 TO 1004 4001 TO 4004 UNI GY -2.8
103. 2001 TO 2004 3001 TO 3004 UNI GY -3.2
104. *STAAD SPACE
105. 5001 TO 5004 6001 TO 6004 UNI GY -2.8
106. LOAD 3 LL
107. MEMBER LOAD
108. 1001 TO 1004 4001 TO 4004 UNI GY -2.1
109. 2001 TO 2004 3001 TO 3004 UNI GY -2.4
110. JOINT LOAD
111. 307 FY -5
112. LOAD 11 WL1 (INTERNAL SUCTION, WIND ANGLE 0)
113. MEMBER LOAD
114. 1001 1002 4001 4002 UNI GY 2.8
115. 1003 1004 4003 4004 UNI GY -1.4
116. 2001 2002 3001 3002 UNI GY 5.16
117. 2003 2004 3003 3004 UNI GY -1.6
118. 201 TO 204 209 TO 212 UNI GZ 1.0
119. 26 28 UNI GZ 5.4
120. 401 TO 404 409 TO 412 UNI GZ 0.2
121. 46 48 UNI GZ 1.0
122. LOAD 12 WL2 (INTERNAL PRESSURE, WIND ANGLE 0)
123. MEMBER LOAD
124. 1001 1002 4001 4002 UNI GY 5.0
125. 1003 1004 4003 4004 UNI GY 1.8
126. 2001 2002 3001 3002 UNI GY 5.7
127. 2003 2004 3003 3004 UNI GY 2.0
128. 201 TO 204 209 TO 212 UNI GZ 0.6
129. 26 28 UNI GZ 3.1
130. 401 TO 404 409 TO 412 UNI GZ 0.6
131. 46 48 UNI GZ 3.1
132. LOAD 13 WL3 (INTERNAL SUCTION, WIND ANGLE 90)
133. MEMBER LOAD
134. 1001 1002 4001 4002 UNI GY -2.2
135. 1003 1004 4003 4004 UNI GY -2.2
136. 2001 2002 3001 3002 UNI GY -2.5
137. 2003 2004 3003 3004 UNI GY -2.5
138. 201 TO 204 209 TO 212 UNI GZ -3.1
139. 301 TO 304 401 TO 404 UNI GX -3.1
140. 301 TO 304 401 TO 404 UNI GX -6.2
141. 309 TO 312 409 TO 412 UNI GX 3.1
142. 209 TO 212 409 TO 412 UNI GX 6.2
143. 309 TO 312 UNI GX 6.2
144. LOAD 14 WL4 (INTERNAL PRESSURE, WIND ANGLE 90)
145. MEMBER LOAD
146. 1001 1002 4001 4002 UNI GY 3.9
147. 1003 1004 4003 4004 UNI GY 3.9
148. 2001 2002 3001 3002 UNI GY 4.5
149. 2003 2004 3003 3004 UNI GY -0.9
150. 201 TO 204 401 TO 404 UNI GX -0.9
151. 201 TO 204 401 TO 404 UNI GX 0.9
152. 209 TO 212 409 TO 412 UNI GX 0.9
153. 309 TO 312 UNI GX 5.0
154. 309 TO 312 UNI GX 5.0
155. LOAD 21 EQ1 (EARTHQUAKE LOAD)
156. JOINT LOAD
157. 206 208 209 406 408 409 FX 15
158. 205 206 208 209 405 406 408 409 FZ 15
159. LOAD 22 EQ1 (EARTHQUAKE LOAD)
160. JOINT LOAD
161. 205 206 208 209 405 406 408 409 FZ 15
```

```
163. LOAD COMB 1001 1.4DL + 1.6LL
164. 1 1.4 2 1.4 3 1.6
166. LOAD COMB 1011 1.2DL + 1.2LL + 1.2WL1
    STAAD SPACE
    --- PAGE NO. 4

167. 1 1.2 2 1.2 3 1.2 11 1.2
168. LOAD COMB 1111 1.4DL + 1.4WL1
169. 1 1.4 2 1.4 11 1.4
170. LOAD COMB 1211 1.0DL + 1.4WL1
171. 1 1.0 2 1.0 11 1.4
172. LOAD COMB 1012 1.2DL + 1.2LL + 1.2WL2
173. 1 1.2 2 1.2 3 1.2 12 1.2
174. LOAD COMB 1112 1.4DL + 1.4WL2
175. 1 1.4 2 1.4 12 1.4
176. LOAD COMB 1212 1.0DL + 1.4WL2
177. 1 1.0 2 1.0 12 1.4
178. LOAD COMB 1013 1.2DL + 1.2LL + 1.2WL3
179. 1 1.2 2 1.2 3 1.2 13 1.2
180. LOAD COMB 1113 1.4DL + 1.4WL3
181. 1 1.4 2 1.4 13 1.4
182. LOAD COMB 1213 1.0DL + 1.4WL3
183. 1 1.0 2 1.0 13 1.4
184. LOAD COMB 1014 1.2DL + 1.2LL + 1.2WL4
185. 1 1.2 2 1.2 3 1.2 14 1.2
186. LOAD COMB 1114 1.4DL + 1.4WL4
187. 1 1.4 2 1.4 14 1.4
188. LOAD COMB 1214 1.0DL + 1.4WL4
189. 1 1.0 2 1.0 14 1.4
191. LOAD COMB 1021 1.2DL + 0.5LL + 1.0EQ1
192. 1 1.2 2 1.2 3 0.5 21 1.0
193. LOAD COMB 1022 1.2DL + 0.5LL + 1.0EQ2
194. 1 1.2 2 1.2 3 0.5 22 1.0
195. LOAD COMB 1121 1.2DL + 0.5LL - 1.0EQ1
196. 1 1.2 2 1.2 3 0.5 21 -1.0
197. LOAD COMB 1122 1.2DL + 0.5LL - 1.0EQ2
198. 1 1.2 2 1.2 3 0.5 22 -1.0
199. LOAD COMB 1221 0.9DL + 1.0EQ1
200. 1 0.9 2 0.9 21 1.0
201. LOAD COMB 1222 0.9DL + 1.0EQ2
202. 1 0.9 2 0.9 22 1.0
203. LOAD COMB 1321 0.9DL - 1.0EQ1
204. 1 0.9 2 0.9 21 -1.0
205. LOAD COMB 1322 0.9DL - 1.0EQ2
206. 1 0.9 2 0.9 22 -1.0
207. *$LS
208. LOAD COMB 2001 1.0DL + 1.0LL
209. 1 1.0 2 1.0 3 1.0
210. LOAD COMB 2002 1.0DL + 0.5LL
211. 1 1.0 2 1.0 3 0.5
212. LOAD COMB 2011 DL + LL + WL1
213. 1 1.0 2 1.0 3 1.0 11 1.0
214. LOAD COMB 2012 DL + LL + WL2
215. 1 1.0 2 1.0 3 1.0 12 1.0
216. LOAD COMB 2013 DL + LL + WL3
217. 1 1.0 2 1.0 3 1.0 13 1.0
218. LOAD COMB 2014 DL + LL + WL4
219. 1 1.0 2 1.0 3 1.0 14 1.0
221. PERFORM ANALYSIS
    STAAD SPACE
    --- PAGE NO. 5

PROBLEM STATISTICS
-----
NUMBER OF JOINTS/MEMBER-ELEMENTS/SUPPORTS = 57/ 86/ 12
ORIGINAL/FINAL BAND-WIDTH= 46/ 14/
TOTAL PRIMARY LOAD CASES = 9, TOTAL DEGREES OF FREEDOM = 306
SIZE OF STIFFNESS MATRIX = 26 DOUBLE KILO-WORDS
REQD/AVAIL. DISK SPACE = 12.5/ 416287.6 MB

**START ITERATION NO. 2
**NOTE-Tension/Compression converged after 2 iterations, Case= 1

**START ITERATION NO. 2
**NOTE-Tension/Compression converged after 2 iterations, Case= 2

**START ITERATION NO. 2
**NOTE-Tension/Compression converged after 2 iterations, Case= 2
```

```
**START ITERATION NO. 3
**NOTE-Tension/Compression converged after 3 iterations, Case= 3
```

```
**START ITERATION NO. 2
**NOTE-Tension/Compression converged after 2 iterations, Case= 11
```

```
**START ITERATION NO. 2
**NOTE-Tension/Compression converged after 2 iterations, Case= 12
```

```
**START ITERATION NO. 2
**NOTE-Tension/Compression converged after 2 iterations, Case= 13
```

```
**START ITERATION NO. 2
**NOTE-Tension/Compression converged after 2 iterations, Case= 14
```

```
**START ITERATION NO. 2
**NOTE-Tension/Compression converged after 2 iterations, Case= 21
```

```
**START ITERATION NO. 2
    STAAD SPACE
    --- PAGE NO. 6
```

```
**NOTE-Tension/Compression converged after 2 iterations, Case= 22
```

```
222. LOAD LIST 1001 1011 TO 1014 1021 1022 1111 TO 1114 1121 1122 -
223. 1211 TO 1214 1221 1222 1321 1322
224. PRINT SUPPORT REACTION ALL
    SUPPORT REACTION ALL
    STAAD SPACE
    --- PAGE NO. 7
```

SUPPORT REACTIONS -UNIT KN NETE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOI-X	MOI-Y	MOI-Z
201	1001	-1.03	63.20	-0.50	0.00	0.00	0.00
	1011	-1.13	32.50	-6.88	0.00	0.00	0.00
	1111	-1.72	26.70	-8.05	0.00	0.00	0.00
	1211	-1.29	12.30	-7.90	0.00	0.00	0.00
	1012	-1.47	22.67	-5.98	0.00	0.00	0.00
	1112	-2.12	15.23	-6.99	0.00	0.00	0.00
	1212	-1.70	0.84	-6.85	0.00	0.00	0.00
	1013	8.32	60.09	-0.43	0.00	0.00	0.00
	1113	9.30	58.89	-0.52	0.00	0.00	0.00
	1213	9.73	44.50	-0.37	0.00	0.00	0.00
	1014	1.10	35.21	-0.47	0.00	0.00	0.00
	1114	0.99	22.86	-0.32	0.00	0.00	0.00
	1214	1.26	12.45	-0.42	0.00	0.00	0.00
	1021	-16.31	70.45	-0.44	0.00	0.00	0.00
	1022	-8.54	33.23	-8.54	0.00	0.00	0.00
	1121	14.08	73.92	-0.44	0.00	0.00	0.00
	1122	-16.17	61.13	7.65	0.00	0.00	0.00
	1221	-0.44	18.43	-8.43	0.00	0.00	0.00
	1321	14.26	59.12	-0.34	0.00	0.00	0.00
	1322	-1.47	46.33	7.76	0.00	0.00	0.00
213	1001	1.03	63.20	-0.50	0.00	0.00	0.00
	1011	1.13	32.50	-6.88	0.00	0.00	0.00
	1111	1.72	26.70	-8.05	0.00	0.00	0.00
	1211	1.29	12.30	-7.90	0.00	0.00	0.00
	1012	1.47	22.67	-5.98	0.00	0.00	0.00
	1112	2.12	15.23	-6.99	0.00	0.00	0.00
	1212	1.70	0.84	-6.85	0.00	0.00	0.00
	1013	-8.32	60.09	-0.43	0.00	0.00	0.00
	1113	-9.30	58.89	-0.52	0.00	0.00	0.00
	1213	-9.73	44.50	-0.37	0.00	0.00	0.00
	1014	-1.10	35.21	-0.47	0.00	0.00	0.00
	1114	-0.99	22.86	-0.32	0.00	0.00	0.00
	1214	-1.26	12.45	-0.42	0.00	0.00	0.00
	1021	16.31	70.45	-0.44	0.00	0.00	0.00
	1022	8.54	33.23	-8.54	0.00	0.00	0.00
	1121	-14.08	73.92	-0.44	0.00	0.00	0.00
	1122	16.17	61.13	7.65	0.00	0.00	0.00
	1221	0.44	18.43	-8.43	0.00	0.00	0.00
	1321	-14.26	59.12	-0.34	0.00	0.00	0.00
	1322	1.47	46.33	7.76	0.00	0.00	0.00
	1001	-1.03	63.20	-0.50	0.00	0.00	0.00
	1011	-1.13	32.50	-6.88	0.00	0.00	0.00
	1111	-1.72	26.70	-8.05	0.00	0.00	0.00
	1211	-1.29	12.30	-7.90	0.00	0.00	0.00
	1012	-1.47	22.67	-5.98	0.00	0.00	0.00
	1112	-2.12	15.23	-6.99	0.00	0.00	0.00
	1212	-1.70	0.84	-6.85	0.00	0.00	0.00

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE									

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MM-M-X	MM-M-Y	MM-M-Z	MY	LD
1	1213	0.38	59.42	0.66	0.00	0.00	0.00	0.00	0.00
	1014	0.14	24.36	0.13	0.00	0.00	0.00	0.00	0.00
	1114	0.05	6.85	-0.17	0.00	0.00	0.00	0.00	0.00
	1214	-0.01	-5.99	-0.30	0.00	0.00	0.00	0.00	0.00
	1021	-5.03	18.08	0.50	0.00	0.00	0.00	0.00	0.00
	1022	0.25	67.29	-9.22	0.00	0.00	0.00	0.00	0.00
	1121	5.46	74.39	0.49	0.00	0.00	0.00	0.00	0.00
	1221	0.19	25.18	10.21	0.00	0.00	0.00	0.00	0.00
	1222	-5.11	0.74	0.29	0.00	0.00	0.00	0.00	0.00
	1321	0.16	49.82	-9.73	0.00	0.00	0.00	0.00	0.00
	1322	5.37	57.84	10.60	0.00	0.00	0.00	0.00	0.00
	1323	0.10	7.84	10.60	0.00	0.00	0.00	0.00	0.00
	1324	0.10	7.84	10.60	0.00	0.00	0.00	0.00	0.00

***** END OF LATEST ANALYSIS RESULT *****

225. PRINT FORCE ENVELOPE NSECTION 5 ALL
FORCE ENVELOPE NSECTION 5
STAAD SPACE

-- PAGE NO. 13

MEMBER FORCE ENVELOPE

ALL UNITS ARE KN METE

MEMB	DISTANCE	FY	LD	MZ	LD	FZ	LD	MY	LD
26	0.00	MAX	9.44	1322	0.00	1022	5.13	1321	0.00
		MIN	-26.18	1111	0.00	1211	-5.48	1021	0.00
	1.17	MAX	9.44	1322	25.41	1111	5.13	1321	5.99
		MIN	-17.36	1111	-11.01	1322	-5.48	1021	-6.40
	2.33	MAX	9.44	1322	40.52	1111	5.13	1321	11.98
		MIN	-10.22	1022	-22.03	1322	-5.48	1021	-12.80
	3.50	MAX	9.44	1322	45.33	1111	5.13	1321	17.97
		MIN	-9.44	1322	-35.04	1322	-5.48	1021	-33.02
	4.67	MAX	9.44	1322	47.70	1322	5.13	1321	51.59
		MIN	-18.07	1211	-49.63	1322	-5.13	1321	-58.95
	5.84	MAX	18.07	1211	-55.07	1322	5.13	1321	-61.99
		MIN	-10.22	1022	-55.07	1322	-5.48	1021	-61.99

MAX/MIN FORCE VALUES FOR MEMB 26, AMONGST ALL SECT LOCATIONS									

MAX.	18.07	5.84	1211	59.63	5.84	1022	80.77	C	0.00
MIN.	-26.18	0.00	1111	-55.07	5.84	1322	35.55	T	5.84
	-5.48	5.84	1021	-31.99	5.84	1021			

28	0.00	MAX	9.44	1322	0.00	1211	5.46	1121	0.00
		MIN	-26.18	1111	0.00	1322	-5.11	1221	0.00
	1.17	MAX	9.44	1322	25.41	1111	5.46	1121	6.37
		MIN	-17.36	1111	-11.01	1322	-5.11	1221	-5.96
	2.33	MAX	9.44	1322	40.52	1111	5.46	1121	12.74
		MIN	-10.22	1022	-22.03	1322	-5.11	1221	-11.93
	3.50	MAX	9.44	1322	45.33	1111	5.46	1121	19.12
		MIN	-9.44	1322	-33.04	1322	-5.11	1221	-33.02
	4.67	MAX	9.44	1322	47.70	1022	5.46	1121	51.59
		MIN	-10.22	1022	-44.05	1322	-5.11	1221	-58.95
	5.84	MAX	18.07	1211	-55.07	1322	5.46	1121	-61.99
		MIN	-10.22	1022	-55.07	1322	-5.11	1221	-61.99

STAAD SPACE

28. AMONGST ALL SECT LOCATIONS
MAX/MIN FORCE VALUES FOR MEMB 28, AMONGST ALL SECT LOCATIONS

-- PAGE NO. 14

Inflam-Mat_Store.ANL									

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MM-M-X	MM-M-Y	MM-M-Z	MY	LD
3006	1321	5.37	57.05	-0.28	0.00	0.00	0.00	0.00	0.00
	1322	0.15	49.49	9.44	0.00	0.00	0.00	0.00	0.00
	1001	-0.22	70.48	-6.67	0.00	0.00	0.00	0.00	0.00
	1111	-0.16	48.76	-7.79	0.00	0.00	0.00	0.00	0.00
	1211	-0.15	33.33	-7.79	0.00	0.00	0.00	0.00	0.00
	1012	-0.06	20.89	-7.79	0.00	0.00	0.00	0.00	0.00
	1112	-0.03	12.16	-7.79	0.00	0.00	0.00	0.00	0.00
	1212	0.02	81.96	-7.79	0.00	0.00	0.00	0.00	0.00
	1013	-0.45	59.62	0.00	0.00	0.00	0.00	0.00	0.00
	1113	-0.15	31.72	0.00	0.00	0.00	0.00	0.00	0.00
	1213	-0.08	12.88	0.00	0.00	0.00	0.00	0.00	0.00
	1014	-0.13	0.43	0.00	0.00	0.00	0.00	0.00	0.00
	1214	-0.13	0.43	0.00	0.00	0.00	0.00	0.00	0.00

-- PAGE NO. 11

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MM-M-X	MM-M-Y	MM-M-Z	MY	LD
3008	1211	0.10	20.89	-7.79	0.00	0.00	0.00	0.00	0.00
	1012	0.06	12.16	-7.79	0.00	0.00	0.00	0.00	0.00
	1112	0.03	6.08	-7.79	0.00	0.00	0.00	0.00	0.00
	1212	-0.02	-0.16	-7.79	0.00	0.00	0.00	0.00	0.00
	1013	0.45	59.62	0.00	0.00	0.00	0.00	0.00	0.00
	1113	0.15	31.72	0.00	0.00	0.00	0.00	0.00	0.00
	1213	0.08	12.88	0.00	0.00	0.00	0.00	0.00	0.00
	1014	0.13	0.43	0.00	0.00	0.00	0.00	0.00	0.00
	1214	0.13	0.43	0.00	0.00	0.00	0.00	0.00	0.00
	1021	-4.65	20.93	0.00	0.00	0.00	0.00	0.00	0.00
	1121	-0.17	46.17	-14.15	0.00	0.00	0.00	0.00	0.00
	1221	4.98	70.58	14.15	0.00	0.00	0.00	0.00	0.00
	1321	0.16	45.33	14.15	0.00	0.00	0.00	0.00	0.00

-- PAGE NO. 12

Page 7

Inflam-Mat_Store.AHL									
	FY/ FZ	DIST LD	DIST LD	MZ/ MY	DIST LD	FX	DIST LD		
MAX.	18.07	5.84 1211	59.63	5.84 1022	80.77	C	0.00 1013		
MIN.	-5.46	0.00 1121	31.86	5.84 1121	35.55	T	5.84 1212		
	-5.11	5.84 1221	-29.82	5.84 1221					
36	0.00	MAX	14.15 1322	0.00 1122	4.69 1321		0.00 1121		
	MIN	14.15 1322	0.00 1211	-4.97 1021	4.69 1321		0.00 1121		
	1.20	MAX	14.15 1322	16.91 1222	4.69 1321		5.61 1321		
	MIN	14.15 1322	-16.91 1322	-4.97 1021	-5.94 1021		-5.61 1321		
	2.39	MAX	14.15 1322	33.82 1222	4.69 1321		11.22 1321		
	MIN	14.15 1322	-33.82 1322	-4.97 1021	-11.89 1021		-11.89 1321		
	3.59	MAX	14.15 1322	50.73 1222	4.69 1321		16.83 1321		
	MIN	14.15 1322	-50.73 1322	-4.97 1021	-17.83 1021		-17.83 1321		
	4.78	MAX	14.15 1322	67.65 1222	4.69 1321		22.43 1321		
	MIN	14.15 1322	-67.65 1322	-4.97 1021	-23.77 1021		-23.77 1321		
	5.97	MAX	14.15 1322	84.56 1222	4.69 1321		28.94 1321		
	MIN	14.15 1322	-84.56 1322	-4.97 1021	-29.71 1021		-29.71 1321		
STAAD SPACE									
MAX/MIN FORCE VALUES FOR MEMB									
	FY/ FZ	DIST LD	DIST LD	MZ/ MY	DIST LD	FX	DIST LD		
MAX.	14.15	0.00 1122	84.56	5.97 1022	81.96	C	0.00 1013		
MIN.	-14.15	0.00 1321	-84.56	5.97 1322					
	-4.97	5.97 1021	-29.71	5.97 1021	4.84	T	5.97 1212		
38	0.00	MAX	14.15 1322	0.00 1222	4.98 1121		0.00 1321		
	MIN	14.15 1322	0.00 1222	-4.97 1021	4.98 1121		0.00 1321		
	1.20	MAX	14.15 1322	16.91 1222	4.98 1121		5.97 1321		
	MIN	14.15 1322	-16.91 1322	-4.97 1021	-5.94 1021		-5.61 1321		
	2.39	MAX	14.15 1322	33.82 1222	4.98 1121		11.22 1321		
	MIN	14.15 1322	-33.82 1322	-4.97 1021	-11.89 1021		-11.89 1321		
	3.59	MAX	14.15 1322	50.73 1222	4.98 1121		16.83 1321		
	MIN	14.15 1322	-50.73 1322	-4.97 1021	-17.83 1021		-17.83 1321		
	4.78	MAX	14.15 1322	67.65 1222	4.98 1121		22.43 1321		
	MIN	14.15 1322	-67.65 1322	-4.97 1021	-23.77 1021		-23.77 1321		
	5.97	MAX	14.15 1322	84.56 1222	4.98 1121		28.94 1321		
	MIN	14.15 1322	-84.56 1322	-4.97 1021	-29.71 1021		-29.71 1321		
STAAD SPACE									
MAX/MIN FORCE VALUES FOR MEMB									
	FY/ FZ	DIST LD	DIST LD	MZ/ MY	DIST LD	FX	DIST LD		
MAX.	14.15	0.00 1122	84.56	5.97 1022	81.96	C	0.00 1013		
MIN.	-14.15	0.00 1321	-84.56	5.97 1322					
	-4.97	5.97 1021	-29.71	5.97 1021	4.84	T	5.97 1212		
46	0.00	MAX	10.21 1122	0.00 1212	5.13 1321		0.00 1221		
	MIN	10.21 1122	0.00 1111	-5.48 1021	5.13 1321		0.00 1221		
	1.17	MAX	10.21 1122	17.50 1222	5.13 1321		5.99 1321		
	MIN	10.21 1122	-17.50 1322	-5.48 1021	-5.94 1021		-5.61 1321		
	2.33	MAX	10.21 1122	34.99 1222	5.13 1321		11.22 1321		
	MIN	10.21 1122	-34.99 1322	-5.48 1021	-11.89 1021		-11.89 1321		
	3.50	MAX	10.21 1122	51.98 1222	5.13 1321		16.83 1321		
	MIN	10.21 1122	-51.98 1322	-5.48 1021	-17.83 1021		-17.83 1321		
	4.67	MAX	10.21 1122	68.97 1222	5.13 1321		22.43 1321		
	MIN	10.21 1122	-68.97 1322	-5.48 1021	-23.77 1021		-23.77 1321		
	5.84	MAX	10.21 1122	85.96 1222	5.13 1321		28.94 1321		
	MIN	10.21 1122	-85.96 1322	-5.48 1021	-29.71 1021		-29.71 1321		
STAAD SPACE									
MAX/MIN FORCE VALUES FOR MEMB									
	FY/ FZ	DIST LD	DIST LD	MZ/ MY	DIST LD	FX	DIST LD		
MAX.	10.21	0.00 1122	85.96	5.13 1022	81.94	C	0.00 1011		
MIN.	-10.21	0.00 1321	-85.96	5.13 1322					
	-5.13	5.13 1021	-29.71	5.13 1021	4.84	T	5.13 1212		

Inflam-Mat_Store.AHL									
	FY/ FZ	DIST LD	DIST LD	MZ/ MY	DIST LD	FX	DIST LD		
MAX.	18.07	5.84 1211	59.63	5.84 1022	80.77	C	0.00 1013		
MIN.	-5.46	0.00 1121	31.86	5.84 1121	35.55	T	5.84 1212		
	-5.11	5.84 1221	-29.82	5.84 1221					
48	0.00	MAX	10.21 1122	0.00 1212	5.46 1121		0.00 1221		
	MIN	10.21 1122	0.00 1111	-5.48 1021	5.46 1121		0.00 1221		
	1.17	MAX	10.21 1122	17.50 1222	5.46 1121		5.99 1321		
	MIN	10.21 1122	-17.50 1322	-5.48 1021	-5.94 1021		-5.61 1321		
	2.33	MAX	10.21 1122	34.99 1222	5.46 1121		11.22 1321		
	MIN	10.21 1122	-34.99 1322	-5.48 1021	-11.89 1021		-11.89 1321		
	3.50	MAX	10.21 1122	51.98 1222	5.46 1121		16.83 1321		
	MIN	10.21 1122	-51.98 1322	-5.48 1021	-17.83 1021		-17.83 1321		
	4.67	MAX	10.21 1122	68.97 1222	5.46 1121		22.43 1321		
	MIN	10.21 1122	-68.97 1322	-5.48 1021	-23.77 1021		-23.77 1321		
	5.84	MAX	10.21 1122	85.96 1222	5.46 1121		28.94 1321		
	MIN	10.21 1122	-85.96 1322	-5.48 1021	-29.71 1021		-29.71 1321		
STAAD SPACE									
MAX/MIN FORCE VALUES FOR MEMB									
	FY/ FZ	DIST LD	DIST LD	MZ/ MY	DIST LD	FX	DIST LD		
MAX.	10.21	0.00 1122	85.96	5.46 1022	81.94	C	0.00 1011		
MIN.	-10.21	0.00 1321	-85.96	5.46 1322					
	-5.11	5.84 1221	-29.82	5.84 1221	10.56	T	5.84 1214		
201	0.00	MAX	20.47 1121	0.00 1321	8.54 1022		0.00 1112		
	MIN	20.47 1121	0.00 1021	-8.54 1022	8.54 1022		0.00 1112		
	0.81	MAX	20.47 1121	12.63 1221	8.54 1022		8.54 1022		
	MIN	20.47 1121	-12.63 1321	-8.54 1022	-8.54 1022		-8.54 1022		
	1.62	MAX	20.47 1121	25.26 1221	8.54 1022		12.63 1321		
	MIN	20.47 1121	-25.26 1321	-8.54 1022	-12.63 1321		-12.63 1321		
	2.43	MAX	20.47 1121	37.89 1221	8.54 1022		19.14 1321		
	MIN	20.47 1121	-37.89 1321	-8.54 1022	-19.14 1321		-19.14 1321		
	3.23	MAX	20.47 1121	50.52 1221	8.54 1022		21.67 1321		
	MIN	20.47 1121	-50.52 1321	-8.54 1022	-21.67 1321		-21.67 1321		
	4.04	MAX	20.47 1121	63.15 1221	8.54 1022		24.20 1321		
	MIN	20.47 1121	-63.15 1321	-8.54 1022	-24.20 1321		-24.20 1321		
STAAD SPACE									
MAX/MIN FORCE VALUES FOR MEMB									
	FY/ FZ	DIST LD	DIST LD	MZ/ MY	DIST LD	FX	DIST LD		
MAX.	20.47	0.00 1121	63.15	4.04 1021	72.41	C	0.00 1121		
MIN.	-20.47	0.00 1021	-63.15	4.04 1022					
	-8.54	4.04 1021	-21.67	4.04 1021	2.17	T	4.04 1212		
	-7.76	4.04 1022	-19.14	4.04 1022					
202	0.00	MAX	13.58 1113	82.05 1121	3.52 1222		6.11 1122		
	MIN	13.58 1113	82.05 1121	-3.52 1021	3.52 1222		6.11 1122		
	0.16	MAX	14.25 1113	82.47 1121	3.52 1222		6.11 1122		
	MIN	14.25 1113	82.47 1121	-3.52 1021	-3.52 1222		-3.52 1222		
	0.33	MAX	14.91 1113	82.88 1121	3.52 1222		6.11 1122		
	MIN	14.91 1113	82.88 1121	-3.52 1021	-3.52 1222		-3.52 1222		
	0.49	MAX	15.58 1113	83.30 1121	3.52 1222		6.11 1122		
	MIN	15.58 1113	83.30 1121	-3.52 1021	-3.52 1222		-3.52 1222		
	0.65	MAX	16.25 1113	83.71 1121	3.52 1222		6.11 1122		
	MIN	16.25 1113	83.71 1121	-3.52 1021	-3.52 1222		-3.52 1222		
	0.82	MAX	16.91 1113	84.12 1121	3.52 1222		6.11 1122		
	MIN	16.91 1113	84.12 1121	-3.52 1021	-3.52 1222		-3.52 1222		
STAAD SPACE									
MAX/MIN FORCE VALUES FOR MEMB									
	FY/ FZ	DIST LD	DIST LD	MZ/ MY	DIST LD	FX	DIST LD		
MAX.	16.91	0.82 1113	84.16	0.82 1121	43.82	C	0.00 1121		
MIN.	-16.91	0.82 1121	-84.16	0.82 1122					
	-4.23	0.82 1121	-74.72	0.82 1121	12.78	T	0.82 1212		
	-4.66	0.82 1122	-72.41	0.82 1122					

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2007	0.00	MAX	0.17	1221	10.85	3013	0.04	1321	0.20	1121
	0.00	MIN	-0.76	1221	11.81	3013	-0.04	1321	-0.43	1221
	0.40	MAX	-0.11	1221	13.01	3013	0.04	1321	0.20	1122
	0.40	MIN	-0.69	1113	11.78	3013	-0.04	1321	-0.43	1211
	0.80	MAX	-0.39	1221	11.33	3013	0.04	1321	0.20	1122
	0.80	MIN	-1.13	1113	11.60	3013	-0.04	1321	-0.43	1211
	1.20	MAX	-0.68	1221	11.79	3013	0.04	1321	0.20	1122
	1.20	MIN	-1.57	1113	12.30	3212	-0.04	1321	-0.43	1211
	1.60	MAX	-0.96	1221	12.40	3013	0.04	1321	0.22	1021
	1.60	MIN	-2.01	1113	12.85	3212	-0.04	1321	-0.43	1021
	2.00	MAX	-1.24	1221	13.16	3013	0.04	1321	0.23	1021
	2.00	MIN	-2.45	1113	13.31	3212	-0.04	1321	-0.43	1211

MAX/MIN FORCE VALUES FOR MEMB						207, AMONGST ALL SECT LOCATIONS						
	FZ	DIST	LD	FY	DIST	LD	MZ	DIST	LD	FX	DIST	LD
MAX.	0.17	0.00	1221	13.16	2.00	1013						
	0.04	0.00	1021	0.23	2.00	1021				2.49	c	2.00 1211
MIN.	-2.45	2.00	1113	-3.84	0.00	1212						
	-0.04	2.00	1013	-0.43	1.60	1211				20.44	x	0.00 1113

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0.00	MAX	36.58	1021	44.59	1221	4.56	1122	4.09	1322
0.05	MAX	37.38	1212	75.01	1221	3.71	1211	-5.50	1222
0.16	MAX	38.38	1212	49.86	1221	4.66	1121	3.51	1222
0.25	MAX	39.72	1021	75.56	1221	-3.52	1222	-4.74	1122
0.33	MAX	43.26	1212	55.16	1221	4.66	1121	2.94	1222
0.49	MAX	39.87	1021	76.47	1221	-3.2	1222	-3.98	1222
0.55	MAX	43.13	1212	60.47	1221	5.06	1121	3.28	1222
0.65	MIN	39.01	1021	68.58	1221	4.26	1121	-2.39	1222
0.75	MIN	40.16	1021	77.02	1121	3.52	1222	-1.70	1222
						-2.45	1122	-2.45	1122

Page 12

Page 12[illegible]

	MAX/MIN FORCE VALUES FOR MEMB			203, AMONGST ALL SECT LOCATIONS					
	FY/ FZ	LD DIST	LD DIST	MZ/ MY	LD DIST	LD DIST	FX	FX DIST	LD DIST
MAX.	31.15	0.82	1013	84.16	0.00	1121			
	3.52	0.00	1222	2.26	0.00	1122	38.38	C	0.00 1121
MIN.	-4.66	0.82	1212	-75.90	0.82	1021			
				-1.81	0.00	1222	15.64	T	0.82 1221

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	MAX/MIN FORCE VALUES FOR MEMB						204, AMONGST ALL SECT LOCATIONS					
	FZ	DIST	LD	FY	DIST	LD	MZ	DIST	LD	FX	DIST	LD
MAX.	35.81	0.00	1121	3.71	0.82	1121	69.60	0.00	1321			
	-13.50	0.82	1122	-4.09	0.82	1222	-77.15	0.82	1021	22.71	C	0.00 1121
MIN.	-4.66	0.82	1122	-5.50	0.82	1122	-5.50	0.82	1122	15.01	T	0.82 1221

2005	0.00	MAX	24.45	1321	46.71	1321	1.56	1322	3.43	1222
		MIN	-35.72	1021	-37.15	1021	-1.13	1211	-4.62	1122
0.63		MAX	24.01	1321	70.56	1321	1.56	1222	2.74	1222
		MIN	-36.50	1021	-41.68	1021	-1.52	1211	-3.65	1122
1.25		MAX	52.86	1321	16.79	1321	1.13	1211	5.67	1122
		MIN	-37.91	1021	-52.75	1021	-1.56	1222	-1.34	1222
1.88		MAX	37.48	1021	8.48	1021	1.13	1211	-1.70	1122
		MIN	-22.68	1021	-15.13	1021	-1.56	1222	0.65	1222
2.50		MAX	38.07	1021	12.24	1321	1.13	1211	-0.72	1122
		MIN	-38.65	1021	-39.23	1021	-1.56	1222	0.45	1021
3.13		MAX	22.24	1321	26.12	1321	1.56	1222	-0.69	1211
		MIN	-38.65	1021	-36.29	1321	-1.13	1211		

MAX/MIN FORCE VALUES FOR MEMB				205, AMONGST ALL SECT LOCATIONS					
	Fy/	LD	DIST	Fy/	LD	DIST	LD	DIST	LD
	Fy/	LD	DIST	Fy/	LD	DIST	LD	FX	DIST
MAX.	24.45	0.00	1321	46.71	0.00	1321			
	1.56	0.00	1122	3.43	0.00	1222			
	-38.05	3.13	1021	-77.15	0.00	1021		3.39	0.00 1222
MIN.	-31.13	3.13	1211	-4.62	0.00	1122			
								21.22	3.13 1113

206	0.00	MAX	2.45	1113	13.16	1013	0.04	1321	0.20	1122
		MIN	1.29	1321	-2.31	1212	-0.04	1321	-0.43	1211
	0.40	MAX	2.01	1113	12.40	1013	0.04	1321	0.20	1122
		MIN	1.01	1321	-2.86	1212	-0.04	1321	-0.43	1211
	0.80	MAX	1.57	1113	11.79	1013	0.04	1321	0.20	1122
		MIN	0.73	1321	-3.30	1212	-0.04	1321	-0.43	1211

7-13

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MAX/MIN FORCE VALUES FOR MEMB						402, AMONGST ALL SECT LOCATIONS					
	FY/ FZ	LD DIST	LD DIST	MV/ DZ	LD DIST		FY DZ	LD DIST	FX DZ	LD DIST	
MAX.	16.91	0.82	1113	84.16	0.82	1121					
	4.79	0.00	1022	4.88	0.00	1322					
	-4.23	0.82	1321	-72.19	0.82	1231			43.82	C 0.00 1121	
MIN.	-3.66	0.82	1322	-6.28	0.00	1022					
									11.49	0.82 1221	

MO3	MAX	29.36	1013	84.16	1121	4.70	1072	1.86	1332
0.00	MIN	-0.51	1214	-2.19	1321	-3.66	1332	-7.30	1072
0.16	MAX	23.42	1013	60.65	1121	1.72	1072	1.72	1072
0.33	MIN	20.08	1214	72.79	1321	-3.66	1332	-1.53	1072
0.49	MAX	30.97	1013	72.79	1121	4.70	1072	0.66	1322
0.65	MIN	30.44	1214	73.18	1321	-3.66	1332	-0.75	1072
0.82	MAX	30.44	1013	74.57	1121	4.70	1072	0.12	1013
0.99	MIN	-0.30	1214	-73.65	1321	-3.66	1332	-0.10	1212
1.16	MAX	30.80	1013	71.88	1121	4.70	1072	0.82	1072
1.33	MIN	-0.23	1214	-74.59	1321	-3.66	1332	-0.53	1322
1.49	MAX	31.15	1013	69.60	1121	4.70	1072	1.60	1072
1.66	MIN	-0.15	1214	-75.90	1321	-3.66	1332	-1.13	1322

MAX/MIN FORCE VALUES FOR MEMB						403, AMONGST ALL SECT LOCATIONS					
	FY/ FZ	DIST LD	DIST LD	MZ/ MY	DIST LD	DIST LD	FX	DIST LD			
MAX.	31.15	0.82	10313		84.16	0.00	11211				
	4.79	0.00	1022		1.86	0.00	1322		38.38 C	0.00 11211	
MIN.	-0.51	0.00	12114		-75.90	0.82	1021				
	-3.66	0.82	1322		-2.31	0.00	1022		15.64 T	0.82 12211	

[illegible]page 18

		Inflan-Mat_Store-ANU		310. AMONGST ALL SECT LOCATIONS	
		70-73	1021	70-73	1021
0.33	MAX	1.27	1212	1.32	1332
	MIN	-30.26	1012	-1.32	1332
0.49	MAX	1.27	1212	1.32	1332
	MIN	-30.45	1013	-1.32	1332
0.66	MAX	1.10	1212	1.32	1332
	MIN	-28.63	1013	-1.32	1332
0.82	MAX	1.02	1212	1.32	1332
	MIN	-27.81	1013	-1.32	1332
STAAD SPACE					
				--	PAGE NO. 27

	MAX.	0.00	MAX	4.88	1221	80.50	0.82	1021	1.32	1322	39.78	C	0.82	1021
	1.44	0.00	1212	-22.79	1213	-70.35	1321	0.67	0.82	1122				
	1.32	0.00	1121	-1.89	1113	68.54	1321	0.67	0.82	1122				
MTN.	-31.91	0.00	1013	-1.91	1013	79.30	1021	-71.95	0.00	1121	22.93	T	0.00	1214
	-1.32	0.82	1212	-4.84	1221	79.30	1021	-0.67	0.82	1222				
				-20.05	1113	-66.84	1321							
				4.82	1221	78.84	09 1321							
	0.49	MAX	4.82	1221	78.84	09 1321								
		MTN	-18.69	1113	-65.09	1021								
	0.66	MAX	4.80	1221	78.30	1021								
		MTN	-17.32	1113	-63.34	1321								
	0.82	MAX	4.78	1221	77.76	1021								
		MTN	-15.95	1113	-61.58	1321								

	MAX/MIN FORCE VALUES FOR MEMB						311, AMONGST ALL SECT LOCATIONS					
	Fx	DIST	Ld	MZ	DIST	Ld	Fy	DIST	Ld	MZ	DIST	Ld
MAX.	4.88	0.00	0.00	1221	80.50	0.00	1021	0.00	0.00	1021	0.00	1021
	1.32	0.00	0.00	1122	1.76	0.82	1122	0.00	0.00	1122	0.00	1122
MIN.	-22.79	0.00	0.00	1113	-70.33	0.00	1321	0.00	0.00	1321	0.00	1321
	-1.32	0.82	1222		-1.76	0.82	1222	16.80	0.00	1321	0.00	1321

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	MAX/MIN FORCE VALUES FOR MEMB						312, AMONGST ALL SECT LOCATIONS					
	Fz	DIST	LD	Fz	DIST	LD	MZ	DIST	LD	Fx	DIST	LD
MAX.	10.91	0.00	1321	61.75	0.00	1321	61.75	0.00	1321			
	14.21	0.00	1122	42.64	0.00	1022						
MIN.	-23.12	4.18	1113	-78.02	0.00	1021				67.09	C	4.18 1021
	-10.21	4.18	1322	-42.64	0.00	1322				2.63	C	0.00 1214

401	0.00	MAX	20.77	1121	0.00	1321	7.77	1222	0.00	1212
		MIN	-13.67	1121	-16.50	1321	-5.77	1122	6.28	1122
	0.51	MAX	19.07	1221	17.52	1121	-4.55	1122	6.28	1122
		MIN	-20.33	1121	-25.53	1221	7.77	1222	12.56	1222
	1.62	MAX	20.33	1221	32.99	1121	-8.55	1122	-13.82	1122
		MIN	-15.72	1221	-38.06	1221	7.77	1222	18.95	1222
	2.43	MAX	20.77	1121	39.40	1121	-8.55	1122	-20.73	1122
		MIN	-15.77	1221	-48.46	1221	7.77	1222	18.95	1222

Page 17

Inflam-Mat_Store.ANL

MAX/MIN FORCE VALUES FOR MEMB										404, AMONGST ALL SECT LOCATIONS									
		FY/		DIST		LD				MZ/		DIST		LD		FX		DIST	
MAX.	MIN.	FZ								MY									
35.81	0.00	MAX	24.45	1321	0.00	1321	69.60	0.00	1321	5.67	0.82	1322	22.71	C	0.00	1321			
4.79	0.00	MIN	0.00	1022	0.00	1022	-77.15	0.82	1021	-77.15	0.82	1021	15.01	T	0.82	1321			
-6.19	0.82		0.82	1322			-4.26	0.82	1322	-4.26	0.82	1322							
-3.66	0.82		0.82	1322															

405										405, AMONGST ALL SECT LOCATIONS									
		FY/		DIST		LD				MZ/		DIST		LD		FX		DIST	
MAX.	MIN.	FZ								MY									
0.00	MAX		24.45	1321	0.00	1321	46.71	1321	1.22	1322	4.80	1022							
0.63	MAX		-35.72	1021	-1.67	1021	-77.15	1021	1.22	1322	-3.61	1322	-1.67	1022	-3.61	1322			
1.25	MAX		24.01	1321	1.22	1322	54.63	1021	1.22	1322	2.84	1322	2.84	1322	2.84	1322			
1.88	MAX		36.89	1021	1.22	1322	31.74	1021	1.22	1322	2.71	1022	2.71	1022	2.71	1022			
2.50	MAX		23.12	1321	1.22	1322	2.08	1321	1.22	1322	1.67	1022	1.67	1022	1.67	1022			
3.13	MAX		36.09	1021	1.22	1322	18.49	1021	1.22	1322	1.67	1022	1.67	1022	1.67	1022			
	MIN		36.09	1021	1.22	1322	13.73	1321	1.22	1322	0.53	1322	0.53	1322	0.53	1322			
	MIN		32.54	1321	1.22	1322	39.12	1021	1.22	1322	0.23	1321	0.23	1321	0.23	1321			
	MIN		-38.65	1021	-1.67	1022	-26.29	1321	-1.67	1022	-0.76	1112	-0.76	1112	-0.76	1112			

406										406, AMONGST ALL SECT LOCATIONS									
		FY/		DIST		LD				MZ/		DIST		LD		FX		DIST	
MAX.	MIN.	FZ								MY									
0.00	MAX		2.45	1113	0.04	1321	13.16	1013	0.04	1321	0.15	1322	0.15	1322	0.15	1322			
0.40	MAX		2.01	1113	0.04	1321	0.91	1214	0.04	1321	-0.49	1112	-0.49	1112	-0.49	1112			
0.80	MAX		1.01	1321	0.04	1321	12.40	1013	0.04	1321	0.15	1322	0.15	1322	0.15	1322			
1.20	MAX		1.57	1113	0.04	1321	11.79	1013	0.04	1321	-0.49	1112	-0.49	1112	-0.49	1112			
1.60	MAX		0.73	1321	0.04	1321	-0.18	1214	0.04	1321	0.15	1322	0.15	1322	0.15	1322			
2.00	MAX		1.13	1113	0.04	1321	11.33	1013	0.04	1321	-0.49	1112	-0.49	1112	-0.49	1112			
	MIN		0.45	1321	-0.04	1321	-0.54	1214	-0.04	1321	-0.49	1112	-0.49	1112	-0.49	1112			
	MIN		0.69	1113	0.04	1321	11.01	1013	0.04	1321	-0.49	1112	-0.49	1112	-0.49	1112			
	MIN		0.16	1321	-0.04	1321	-0.78	1214	-0.04	1321	-0.49	1112	-0.49	1112	-0.49	1112			
	MIN		0.26	1113	0.04	1321	10.85	1013	0.04	1321	0.15	1322	0.15	1322	0.15	1322			
	MIN		-0.12	1321	-0.04	1321	-0.89	1214	-0.04	1321	-0.49	1112	-0.49	1112	-0.49	1112			

407										407, AMONGST ALL SECT LOCATIONS									
		FY/		DIST		LD				MZ/		DIST		LD		FX		DIST	
MAX.	MIN.	FZ								MY									
0.00	MAX		0.17	1221	0.04	1321	10.85	1013	0.04	1321	0.15	1322	0.15	1322	0.15	1322			
0.40	MAX		-0.16	1113	-0.04	1321	-0.89	1214	-0.04	1321	-0.49	1112	-0.49	1112	-0.49	1112			
0.80	MAX		-0.11	1221	0.04	1321	11.01	1013	0.04	1321	-0.49	1112	-0.49	1112	-0.49	1112			
1.20	MAX		-0.09	1321	0.04	1321	0.38	1214	0.04	1321	0.15	1322	0.15	1322	0.15	1322			
1.60	MAX		-1.33	1113	-0.04	1321	-0.54	1214	-0.04	1321	-0.49	1112	-0.49	1112	-0.49	1112			
2.00	MAX		-0.68	1221	-0.04	1321	11.79	1013	0.04	1321	-0.49	1112	-0.49	1112	-0.49	1112			
	MIN		-1.57	1113	-0.04	1321	-0.18	1214	-0.04	1321	-0.49	1112	-0.49	1112	-0.49	1112			
	MIN		-0.96	1221	-0.04	1321	12.40	1013	0.04	1321	0.15	1322	0.15	1322	0.15	1322			
	MIN		-2.01	1113	-0.04	1321	0.30	1214	-0.04	1321	-0.49	1112	-0.49	1112	-0.49	1112			
	MIN		-1.24	1221	-0.04	1321	13.16	1013	0.04	1321	0.15	1322	0.15	1322	0.15	1322			
	MIN		-2.45	1113	-0.04	1321	0.91	1214	-0.04	1321	-0.49	1112	-0.49	1112	-0.49	1112			

408										408, AMONGST ALL SECT LOCATIONS									
		FY/		DIST		LD				MZ/		DIST		LD		FX		DIST	
MAX.	MIN.	FZ								MY									
0.00	MAX		37.78	1121	0.00	1321	44.58	3.13	1221	4.80	3.13	1022	5.68	C	3.13	1221			
0.63	MAX		1.67	0.00	1022	0.00	1022	4.80	3.13	1221	-75.02	3.13	1121	21.22	T	0.00	1113		
1.25	MAX		-23.57	3.13	1322	3.13	1322	-3.61	3.13	1322	-3.61	3.13	1322						
1.88	MAX		-1.22	3.13	1322	3.13	1322												

409										409, AMONGST ALL SECT LOCATIONS									
		FY/		DIST		LD				MZ/		DIST		LD		FX		DIST	
MAX.	MIN.	FZ								MY									
0.00	MAX		10.69	1321	0.00	1321	71.15	0.82	1221	5.67	0.00	1022	28.86	C	0.82	1021			
0.16	MAX		3.66	0.00	1322	0.00	1322	5.67	0.00	1022	-77.46	0.82	1121	21.17	T	0.00	1321		
0.33	MAX		-40.31	0.82	1022	0.82	1022	-4.26	0.00	1322	-4.26	0.00	1322						
0.49	MAX		0.00	1321	0.15	1214	71.15	1221	3.66	1322	3.66	1322	3.66	1322	3.66	1322			
0.65	MAX		-31.15	1013	-31.15	1013	73.55	1221	3.66	1322	3.66	1322	3.66	1322	3.66	1322			
0.82	MAX		0.23	1214	0.23	1214	76.36	1121	3.66	1322	3.66	1322	3.66	1322	3.66	1322			
	MIN		0.30	1214	0.30	1214	76.36	1121	3.66	1322	3.66	1322	3.66	1322	3.66	1322			
	MIN		-30.44	1013	-30.44	1013	75.44	1221	3.66	1322	3.66	1322	3.66	1322	3.66	1322			
	MIN		0.15	1214	0.15	1214	73.55	1221	3.66	1322	3.66	1322	3.66	1322	3.66	1322			
	MIN		-0.49	1112	-0.49	1112	72.99	1321	3.66	1322	3.66	1322	3.66	1322	3.66	1322			
	MIN		-0.49	1112	-0.49	1112	74.72	1321	3.66	1322	3.66	1322	3.66	1322	3.66	1322			
	MIN		0.51	1214	0.51	1214	86.31	1021	3.66	1322	3.66	1322	3.66	1322	3.66	1322			
	MIN		-29.36	1013	-29.36	1013	74.33	1321	3.66	1322	3.66	1322	3.66	1322	3.66	1322			

407, AMONGST ALL SECT LOCATIONS																	
MAX/MIN FORCE VALUES FOR MEMB					LD												
FY/		DIST		LD		MZ/		DIST		LD		FX		DIST		LD	
FZ		DIST		LD		MY		DIST		LD							
Page 19																	

						Inflam-Wat_Store	ANL		
1.00	MAX	15.98	1013	-6.97	1212	0.18	1122	0.29	1222
	MN	-7.49	1212	-2.06	1214	-0.15	1222	-0.37	1122
2.00	NAX	7.09	1013	3.79	1212	0.18	1122	0.14	1222
	MN	-3.61	1212	-4.58	1212	-0.15	1222	-0.19	1122
3.00	MAX	0.57	1214	5.45	1212	0.18	1122	0.02	1321
	MN	-1.80	1013	-7.23	1013	-0.15	1222	-0.04	1021
4.00	MAX	4.16	1212	3.87	1212	0.18	1122	0.18	1122
	MN	-10.70	1013	-0.98	1013	-0.15	1222	-0.16	1222
5.00	MAX	8.05	1212	14.18	1013	0.18	1122	0.36	1122
	MN	-19.60	1013	-4.35	1214	-0.15	1222	0.31	1222
					STAAD SPACE	--	PAGE NO.	135	

MAX/MIN FORCE VALUES FOR MEMB				1002, AMONGST ALL SECT LOCATIONS			
	FY/ FZ	LD DIST	LD DIST	MZ/ MY	LD DIST	LD DIST	LD DIST
MAX.	24.89	0.00	1013	27.42	0.00	1013	
	0.18	0.00	1132	0.44	0.00	1222	21.37 C
MIN.	-19.60	5.00	1013	-11.20	0.00	1212	21.81 Y
	-0.15	5.00	1022	-0.55	0.00	1122	5.00 1121

1003	0.00	MAX	19.60	1013	14.18	1013	0.14	1322	0.35	1022
		MIN	-5.76	1214	-0.18	1022	0.18	1022	-0.30	1022
1.00		MAX	10.13	1013	0.18	1022	0.14	1322	-0.17	1022
		MIN	-2.10	1214	-0.35	1111	0.18	1022	-0.16	1322
2.00		MAX	1.30	1013	-1.46	1214	0.14	1322	0.02	1322
		MIN	-1.87	1214	-8.90	1011	0.18	1022	-0.04	1021
3.00		MAX	1.77	1214	0.88	1214	0.14	1322	0.13	1322
		MIN	-7.09	1013	-6.00	1011	0.18	1022	-0.18	1022
4.00		MAX	4.12	1214	6.97	1013	0.14	1322	-0.27	1322
		MIN	-15.99	1013	-2.06	1214	-0.18	1022	-0.35	1022
5.00		MAX	6.46	1214	27.42	1013	0.14	1322	-0.42	1322
		MIN	-24.89	1013	-7.36	1214	-0.18	1022	-0.53	1022

MAX/MIN FORCE VALUES FOR MEMB						1003, AMONGST ALL SECT LOCATIONS							
	FZ/ FZ	DIST LD	DIST LD	MZ/ MZ	DIST LD	DIST LD	FX	DIST LD					
MAX.	19.60	0.00	1013	27.42	5.00	1013							
	0.14	0.00	1322	0.42	5.00	1322							
MIN.	-24.89	5.00	1013	-8.90	2.00	1011	21.37	C	5.00	1021			
	-0.18	5.00	1022	-0.53	5.00	1022	21.81	T	0.00	1121			

1.00	MAX	0.00	22.528	1013	27.82	1013	0.00	1321	0.00	1021
1.00	MIN	0.50	1.80	1013	17.80	1013	0.00	1321	0.00	1021
1.00	MAX	1.00	4.69	1214	4.69	1214	0.00	1321	0.00	1021
1.00	MIN	1.00	13.35	1013	10.01	1013	0.00	1321	0.00	1021
1.50	MAX	1.50	8.90	1013	4.15	1013	0.00	1321	0.00	1021
1.50	MIN	2.00	2.34	1214	1.17	1214	0.00	1321	0.00	1021
2.00	MAX	2.00	4.47	1013	1.23	1013	0.00	1013	0.00	1321
2.00	MIN	2.50	7.10	1212	7.10	1212	0.00	1013	0.00	1321
2.50	MAX	2.50	0.00	1112	0.00	1021	0.00	1321	0.00	1111
2.50	MIN		0.00	1112	0.00	1021	0.00	1321	0.00	1112
							---	PAGE NO.	36	

	MAX/MIN FORCE VALUES FOR MEMB						1004, AMONGST ALL SECT LOCATIONS					
	Fy/ Fz	LD DIST	LD DIST	LD DIST	MZ/ MY	LD DIST	LD DIST	LD DIST	LD DIST	FX	DIST	LD DIST
MAX.	22.35	0.00	1013		27.82	0.00	1013					
	0.00	0.00	1321		0.00	0.00	1021					
	-5.86	0.00	1214		-7.33	0.00	1321			0.16	C	0.00 1214
MIN.	0.00	0.00	1021		0.00	0.00	1314			0.62	T	0.00 1013

	0.00	MAX	0.00	1212	0.00	1212	0.00	1021	0.00	1321
20001		MIN	0.00	1222	0.00	1111	0.00	1021	0.00	1021
	0.50	MAX	-2.23	1212	0.00	1013	0.00	1321	0.00	1222
		MIN	-5.05	1013	-0.56	1212	0.00	1321	0.00	1212
	1.00	MAX	4.46	1212	5.05	1013	0.00	1021	0.00	1222
		MIN	-10.10	1013	-2.23	1212	0.00	1321	0.00	1122

Page 22

	-4.79	0.82	1022	-2.31	0.82	1022	73	23	T	0.00	1321
0.00	MAX	7.48	1221	86.31	1021	3.66	1322	1.89	1322		
0.16	MIN	-16.91	1113	-74.33	1321	-4.79	1022	-2.36	1022		
	MIN	7.46	1221	85.34	1321	3.66	1322	2.49	1322		
0.33	MIN	-16.25	1113	-72.10	1321	-4.79	1022	-3.05	1022		
	MIN	7.44	1221	84.39	1021	3.66	1322	3.09	1322		
0.49	MIN	-17.35	1113	-99.46	1021	-3.62	1322	-3.93	1322		
	MIN	7.45	1221	85.34	1321	3.66	1322	4.71	1022		
0.65	MIN	-15.34	1121	-67.63	1321	-4.79	1022	-4.79	1022		
	MIN	7.40	1221	82.49	1021	3.66	1322	4.29	1322		
0.82	MIN	-15.36	1121	-65.37	1321	-4.79	1022	-5.49	1022		
	MIN	7.38	1221	81.54	1021	3.66	1322	4.88	1322		
	MIN	-15.39	1121	-63.12	1321	-4.79	1022	-6.28	1022		

MAX/MIN FORCE VALUES FOR MEMB						411. AMONGST ALL SECT LOCATIONS					
FY/	DIST	LD	FY/	DIST	LD	MZ/	DIST	LD	FX	DIST	LD
MAX.	7.48	0.00	1221			86.31	0.00	1021			
	3.66	0.00	1322			4.88	0.82	1322			
MIN.	-16.91	0.00	1113			-74.33	0.00	1321			
	-4.79	0.82	1022			-6.28	0.82	1022			
									50.72	C	0.82 1021
									18.39	T	0.00 1321

M12	0.00	MAX	15.74	1321	63.12	1321	8.55	1122	31.41	1222
		MIN	-20.01	1021	-81.56	1021	-7.77	1222	-34.55	1222
	0.81	MAX	15.69	1321	50.42	1321	8.55	1222	25.13	1222
		MIN	-20.08	1021	-65.35	1021	-7.77	1222	-27.64	1222
	1.62	MAX	15.64	1321	37.75	1321	8.55	1222	18.85	1222
		MIN	-20.14	1021	-49.09	1021	-7.77	1222	-20.73	1222
	2.43	MAX	15.59	1321	25.13	1321	8.55	1222	12.56	1222
		MIN	-20.21	1021	-32.78	1021	-7.77	1222	-13.82	1222
	3.23	MAX	15.57	1321	12.43	1321	8.55	1222	6.28	1222
		MIN	-20.27	1021	-16.43	1021	-7.77	1222	-6.28	1222
	4.04	MAX	15.49	1321	0.00	1321	8.55	1222	0.00	1112
		MIN	-20.34	1021	0.00	1021	-7.77	1222	0.00	1222
STAAD SPACE										
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	MAX/MIN FORCE VALUES FOR MEMB						412, AMONGST ALL SECT LOCATIONS					
	FY/ DIST	L/D	FY/ DIST	L/D	MZ/ DIST	L/D	FY/ DIST	L/D	MZ/ DIST	L/D	PX DIST	L/D
MAX.	15.74	0.00	1321				63.12	0.00	1321			
	8.55	0.00	1122				31.41	0.00	1272			
MTN.	-20.34	4.04	1021				-81.56	0.00	1021		73.49 C	4.04 1021
	-7.77	4.04	1222				-34.55	0.00	1122		3.12 C	0.00 1321

0001	0.00	MAX	0.00	1021	0.00	1021	0.00	1221	0.00	1222	0.00	1221
	MIN		0.00	1021		1021		1221		1222		1221
	0.50	MAX	1.34	1212		1212		1021		0.00		0.00
	MIN		-3.88	1015		1015		1212		0.00		0.00
	1.00	MAX	-8.90	1013		1013		1215		0.00		0.00
	MIN		-8.90	1013		1013		1215		0.00		0.00
	1.50	MAX	5.83	1013		1013		1013		0.00		0.00
	MIN		-13.35	1012		1012		1012		0.00		0.00
	2.00	MAX	7.77	1212		1212		1013		0.00		0.00
	MIN		-17.60	1013		1013		1212		0.00		0.00
	2.50	MAX	9.71	1212		1212		1212		0.00		0.00
	MIN		-22.25	1013		1013		1212		0.00		0.00
								1212		0.00		0.00

	MAX/MIN FORCE VALUES FOR MEMB				1001, AMONGST ALL SECT LOCATIONS			
	FY/ FZ	DIST LD	DIST LD	NY/ MZ	DIST LD	DIST LD	FX	DIST LD
MAX.	9.71	2.50 1212		27.82	2.50 1013			
	0.00	0.00 1222		0.00	2.50 1222			
MIN.	-22.25	2.50 1013		-12.14	2.50 1212		0.27 C	2.50 1212
	0.00	2.50 1122		0.00	2.50 1122		0.62 T	2.50 1013

0002	0.00	24.89	1013	27.42	1013	0.18	1122	0.44	1222
		MIN		-11.20	1212	-0.15	1222	-0.55	1122

Page 21

12 2200 21

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MAX/MIN FORCE VALUES FOR MEMB 2004, AMONGST ALL SECT LOCATIONS									
FY/ FZ	DIST LD	DIST LD	DIST LD	DIST LD	DIST LD	DIST LD	DIST LD	FX	DIST LD
MAX.	25.25	0.00	0.00	0.00	0.00	0.00	0.00	0.19	C
MIN.	-6.96	0.00	0.00	0.00	0.00	0.00	0.00	0.70	T

3001	0.00	MAX	0.00	1122	0.00	1122	0.00	1021	0.00	1321
	0.50	MIN	0.00	1222	0.00	1222	0.00	1321	0.00	1021
	1.00	MIN	0.00	1013	0.00	1013	0.00	1021	0.00	1021
	1.50	MIN	0.00	1013	0.00	1013	0.00	1021	0.00	1021
	2.00	MIN	0.00	1013	0.00	1013	0.00	1021	0.00	1021
	2.50	MIN	0.00	1013	0.00	1013	0.00	1021	0.00	1021

MAX/MIN FORCE VALUES FOR MEMB 3001, AMONGST ALL SECT LOCATIONS									
FY/ FZ	DIST LD	DIST LD	DIST LD	DIST LD	DIST LD	DIST LD	DIST LD	FX	DIST LD
MAX.	11.16	0.00	0.00	0.00	0.00	0.00	0.00	0.31	C
MIN.	-25.25	0.00	0.00	0.00	0.00	0.00	0.00	0.70	T

3002	0.00	MAX	0.00	1122	0.00	1122	0.00	1021	0.00	1321
	1.00	MIN	0.00	1222	0.00	1222	0.00	1021	0.00	1021
	2.00	MIN	0.00	1013	0.00	1013	0.00	1021	0.00	1021
	3.00	MIN	0.00	1013	0.00	1013	0.00	1021	0.00	1021
	4.00	MIN	0.00	1013	0.00	1013	0.00	1021	0.00	1021
	5.00	MIN	0.00	1013	0.00	1013	0.00	1021	0.00	1021

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MAX/MIN FORCE VALUES FOR MEMB 3002, AMONGST ALL SECT LOCATIONS									
FY/ FZ	DIST LD	DIST LD	DIST LD	DIST LD	DIST LD	DIST LD	DIST LD	FX	DIST LD
MAX.	34.42	0.00	0.00	0.00	0.00	0.00	0.00	16.71	C
MIN.	-32.97	0.00	0.00	0.00	0.00	0.00	0.00	8.49	T

3003	0.00	MAX	0.00	1122	0.00	1122	0.00	1021	0.00	1321
	1.00	MIN	0.00	1222	0.00	1222	0.00	1021	0.00	1021
	2.00	MIN	0.00	1013	0.00	1013	0.00	1021	0.00	1021
	3.00	MIN	0.00	1013	0.00	1013	0.00	1021	0.00	1021
	4.00	MIN	0.00	1013	0.00	1013	0.00	1021	0.00	1021
	5.00	MIN	0.00	1013	0.00	1013	0.00	1021	0.00	1021

MAX/MIN FORCE VALUES FOR MEMB 3003, AMONGST ALL SECT LOCATIONS									
FY/ FZ	DIST LD	DIST LD	DIST LD	DIST LD	DIST LD	DIST LD	DIST LD	FX	DIST LD
MAX.	34.42	0.00	0.00	0.00	0.00	0.00	0.00	16.71	C
MIN.	-32.97	0.00	0.00	0.00	0.00	0.00	0.00	8.49	T

Inflam-Mat_Store.ANL									
1.50	MAX	6.70	1212	0.00	1021	0.00	1221	0.00	1221
2.00	MIN	-15.15	1013	0.00	1321	0.00	1021	0.00	1021
2.50	MIN	-20.20	1013	0.00	1021	0.00	1021	0.00	1021
	MAX	-20.20	1013	0.00	1021	0.00	1021	0.00	1021
	MIN	-25.25	1013	0.00	1021	0.00	1021	0.00	1021

MAX/MIN FORCE VALUES FOR MEMB 2001, AMONGST ALL SECT LOCATIONS									
FY/ FZ	DIST LD	DIST LD	DIST LD	DIST LD	DIST LD	DIST LD	DIST LD	FX	DIST LD
MAX.	11.16	0.00	0.00	0.00	0.00	0.00	0.00	0.31	C
MIN.	-25.25	0.00	0.00	0.00	0.00	0.00	0.00	0.70	T

2002	0.00	MAX	0.00	1122	0.00	1122	0.00	1021	0.00	1321
	1.00	MIN	0.00	1222	0.00	1222	0.00	1021	0.00	1021
	2.00	MIN	0.00	1013	0.00	1013	0.00	1021	0.00	1021
	3.00	MIN	0.00	1013	0.00	1013	0.00	1021	0.00	1021
	4.00	MIN	0.00	1013	0.00	1013	0.00	1021	0.00	1021
	5.00	MIN	0.00	1013	0.00	1013	0.00	1021	0.00	1021

?

MAX/MIN FORCE VALUES FOR MEMB 2002, AMONGST ALL SECT LOCATIONS									
FY/ FZ	DIST LD	DIST LD	DIST LD	DIST LD	DIST LD	DIST LD	DIST LD	FX	DIST LD
MAX.	34.42	0.00	0.00	0.00	0.00	0.00	0.00	16.71	C
MIN.	-32.97	0.00	0.00	0.00	0.00	0.00	0.00	8.32	T

2003	0.00	MAX	0.00	1122	0.00	1122	0.00	1021	0.00	1321
	1.00	MIN	0.00	1222	0.00	1222	0.00	1021	0.00	1021
	2.00	MIN	0.00	1013	0.00	1013	0.00	1021	0.00	1021
	3.00	MIN	0.00	1013	0.00	1013	0.00	1021	0.00	1021
	4.00	MIN	0.00	1013	0.00	1013	0.00	1021	0.00	1021
	5.00	MIN	0.00	1013	0.00	1013	0.00	1021	0.00	1021

MAX/MIN FORCE VALUES FOR MEMB 2003, AMONGST ALL SECT LOCATIONS									
FY/ FZ	DIST LD	DIST LD	DIST LD	DIST LD	DIST LD	DIST LD	DIST LD	FX	DIST LD
MAX.	32.96	0.00	0.00	0.00	0.00	0.00	0.00	7.90	C
MIN.	-35.78	0.00	0.00	0.00	0.00	0.00	0.00	8.32	T

2004	0.00	MAX	0.00	1122	0.00	1122	0.00	1021	0.00	1321
	0.50	MIN	0.00	1222	0.00	1222	0.00	1021	0.00	1021
	1.00	MIN	0.00	1013	0.00	1013	0.00	1021	0.00	1021
	1.50	MIN	0.00	1013	0.00	1013	0.00	1021	0.00	1021
	2.00	MIN	0.00	1013	0.00	1013	0.00	1021	0.00	1021
	2.50	MIN	0.00	1013	0.00	1013	0.00	1021	0.00	1021

2.50 MAX 0.00 1122 0.00 1122 0.00 1122 0.00 1211 0.00 1211
 MIN 0.00 1122 0.00 1222 0.00 1222 0.00 1321 0.00 1322 0.00 1322
 STAAD SPACE

Inflam-Mat_Store.ANL

MAX/MIN FORCE VALUES FOR MEMB									
FY/ FZ		DIST LD		DIST LD		DIST LD		DIST LD	
MAX.	0.30	0.00	1001	0.00	0.00	1001	0.00	0.00	1001
MIN.	-0.30	0.00	1114	0.00	5.44	1322	0.00	5.44	1322

0.00 1021 0.00 1122 0.00 1122 0.00 1211 0.00 1211
 0.00 1321 0.00 1222 0.00 1222 0.00 1321 0.00 1322 0.00 1322
 STAAD SPACE

MAX/MIN FORCE VALUES FOR MEMB									
FY/ FZ		DIST LD		DIST LD		DIST LD		DIST LD	
MAX.	0.00	0.00	1122	0.00	2.50	1021	0.00	0.00	1001
MIN.	-12.54	0.00	1114	0.00	2.50	1122	0.00	0.00	1321

MAX/MIN FORCE VALUES FOR MEMB									
FY/ FZ		DIST LD		DIST LD		DIST LD		DIST LD	
MAX.	0.30	0.00	1001	0.00	0.00	1001	0.00	0.00	1001
MIN.	-0.30	0.00	1114	0.00	5.44	1322	0.00	5.44	1322

MAX/MIN FORCE VALUES FOR MEMB									
FY/ FZ		DIST LD		DIST LD		DIST LD		DIST LD	
MAX.	0.00	0.00	1122	0.00	2.50	1021	0.00	0.00	1001
MIN.	-12.54	0.00	1114	0.00	2.50	1122	0.00	0.00	1321

MAX/MIN FORCE VALUES FOR MEMB									
FY/ FZ		DIST LD		DIST LD		DIST LD		DIST LD	
MAX.	0.00	0.00	1122	0.00	2.50	1021	0.00	0.00	1001
MIN.	-12.54	0.00	1114	0.00	2.50	1122	0.00	0.00	1321

STAAD SPACE

--- PAGE NO. 48

MAX/MIN FORCE VALUES FOR MEMB									
FY/ FZ		DIST LD		DIST LD		DIST LD		DIST LD	
MAX.	0.00	0.00	1001	0.00	0.00	1001	0.00	0.00	1001
MIN.	0.00	5.51	1322	0.00	5.51	1322	0.00	5.51	1322

MAX/MIN FORCE VALUES FOR MEMB									
FY/ FZ		DIST LD		DIST LD		DIST LD		DIST LD	
MAX.	0.00	0.00	1001	0.00	0.00	1001	0.00	0.00	1001
MIN.	0.00	5.51	1322	0.00	5.51	1322	0.00	5.51	1322

MAX/MIN FORCE VALUES FOR MEMB									
FY/ FZ		DIST LD		DIST LD		DIST LD		DIST LD	
MAX.	0.00	0.00	1114	0.00	0.00	1114	0.00	0.00	1114
MIN.	-0.30	0.00	1122	0.00	5.51	1322	0.00	5.51	1322

MAX/MIN FORCE VALUES FOR MEMB									
FY/ FZ		DIST LD		DIST LD		DIST LD		DIST LD	
MAX.	0.00	0.00	1114	0.00	0.00	1114	0.00	0.00	1114
MIN.	-0.30	0.00	1122	0.00	5.51	1322	0.00	5.51	1322

STAAD SPACE

--- PAGE NO. 47

MAX/MIN FORCE VALUES FOR MEMB									
FY/ FZ		DIST LD		DIST LD		DIST LD		DIST LD	
MAX.	0.30	0.00	1001	0.00	0.00	1001	0.00	0.00	1001
MIN.	-0.30	0.00	1114	0.00	5.44	1322	0.00	5.44	1322

MAX/MIN FORCE VALUES FOR MEMB									
FY/ FZ		DIST LD		DIST LD		DIST LD		DIST LD	
MAX.	0.30	0.00	1001	0.00	0.00	1001	0.00	0.00	1001
MIN.	-0.30	0.00	1114	0.00	5.44	1322	0.00	5.44	1322

MAX/MIN FORCE VALUES FOR MEMB									
FY/ FZ		DIST LD		DIST LD		DIST LD		DIST LD	
MAX.	0.00	0.00	1114	0.00	0.00	1114	0.00	0.00	1114
MIN.	-0.30	0.00	1122	0.00	5.51	1322	0.00	5.51	1322

MAX/MIN FORCE VALUES FOR MEMB									
FY/ FZ		DIST LD		DIST LD		DIST LD		DIST LD	
MAX.	0.00	0.00	1114	0.00	0.00	1114	0.00	0.00	1114
MIN.	-0.30	0.00	1122	0.00	5.51	1322	0.00	5.51	1322

STAAD SPACE

--- PAGE NO. 49

MAX/MIN FORCE VALUES FOR MEMB									
FY/ FZ		DIST LD		DIST LD		DIST LD		DIST LD	
MAX.	0.00	0.00	1114	0.00	0.00	1114	0.00	0.00	1114
MIN.	-0.30	0.00	1122	0.00	5.51	1322	0.00	5.51	1322

MAX/MIN FORCE VALUES FOR MEMB									
FY/ FZ		DIST LD		DIST LD		DIST LD		DIST LD	
MAX.	0.00	0.00	1114	0.00	0.00	1114	0.00	0.00	1114
MIN.	-0.30	0.00	1122	0.00	5.51	1322	0.00	5.51	1322

Page 29

Page 30

1.18 MAX 0.00 1322 0.00 1322 0.00 1322
MIN 0.00 1322 0.00 1322 0.00 1322
2.36 MAX 0.00 1322 0.00 1322 0.00 1322
MIN 0.00 1322 0.00 1322 0.00 1322
3.54 MAX 0.00 1322 0.00 1322 0.00 1322
MIN 0.00 1322 0.00 1322 0.00 1322
4.72 MAX 0.00 1322 0.00 1322 0.00 1322
MIN 0.00 1322 0.00 1322 0.00 1322
5.90 MAX 0.00 1322 0.00 1322 0.00 1322
MIN 0.00 1322 0.00 1322 0.00 1322

4.72 MAX 0.00 1322 0.00 1322 0.00 1322
MIN 0.00 1322 0.00 1322 0.00 1322
5.90 MAX 0.00 1322 0.00 1322 0.00 1322
MIN 0.00 1322 0.00 1322 0.00 1322

STAAD SPACE

--- PAGE NO. 54

MAX/MIN FORCE VALUES FOR MEMB 8009, AMONGST ALL SECT LOCATIONS									
FY/	FZ/	DIST	LD	MZ/	DIST	LD	FX	DIST	LD
MAX.	0.00	0.00	1001	0.00	0.00	1001	3.46 C	0.00	1122
MIN.	0.00	0.00	1001	0.00	0.00	1001	3.46 T	5.90	1222

8010	0.00	MAX	0.34	1114	0.00	1322	0.00	1322	0.00	1322
	1.18	MIN	0.22	1322	0.00	1322	0.00	1322	0.00	1322
	2.36	MAX	0.00	1322	0.00	1322	0.00	1322	0.00	1322
	3.54	MIN	0.00	1322	0.00	1322	0.00	1322	0.00	1322
	4.72	MAX	0.00	1322	0.00	1322	0.00	1322	0.00	1322
	5.90	MIN	-0.22	1322	0.00	1322	0.00	1322	0.00	1322

STAAD SPACE

--- PAGE NO. 53

MAX/MIN FORCE VALUES FOR MEMB 8010, AMONGST ALL SECT LOCATIONS									
FY/	FZ/	DIST	LD	MZ/	DIST	LD	FX	DIST	LD
MAX.	0.34	0.00	1001	0.00	0.00	1001	8.14 C	5.90	1321
MIN.	-0.34	0.00	1114	0.00	5.90	1322	9.23 T	0.00	1021

8011	0.00	MAX	0.34	1114	0.00	1322	0.00	1322	0.00	1322
	1.18	MIN	0.22	1322	0.00	1322	0.00	1322	0.00	1322
	2.36	MAX	0.00	1322	0.00	1322	0.00	1322	0.00	1322
	3.54	MIN	0.00	1322	0.00	1322	0.00	1322	0.00	1322
	4.72	MAX	0.00	1322	0.00	1322	0.00	1322	0.00	1322
	5.90	MIN	-0.22	1322	0.00	1322	0.00	1322	0.00	1322

MAX/MIN FORCE VALUES FOR MEMB 8011, AMONGST ALL SECT LOCATIONS									
FY/	FZ/	DIST	LD	MZ/	DIST	LD	FX	DIST	LD
MAX.	0.34	0.00	1001	0.00	0.00	1001	8.14 C	5.90	1321
MIN.	-0.34	0.00	1114	0.00	5.90	1322	9.23 T	0.00	1021

8012	0.00	MAX	0.00	1322	0.00	1322	0.00	1322	0.00	1322
	1.18	MIN	0.00	1322	0.00	1322	0.00	1322	0.00	1322
	2.36	MAX	0.00	1322	0.00	1322	0.00	1322	0.00	1322
	3.54	MIN	0.00	1322	0.00	1322	0.00	1322	0.00	1322

Page 33

MAX/MIN FORCE VALUES FOR MEMB 8012, AMONGST ALL SECT LOCATIONS									
FY/	FZ/	DIST	LD	MZ/	DIST	LD	FX	DIST	LD
MAX.	0.00	0.00	1001	0.00	0.00	1001	0.00	0.00	1001
MIN.	0.00	0.00	1001	0.00	5.90	1322	4.44 T	5.90	1212

9005	0.00	MAX	0.00	1322	0.00	1322	0.00	1322	0.00	1322
	1.10	MIN	0.00	1322	0.00	1322	0.00	1322	0.00	1322
	2.21	MAX	0.00	1322	0.00	1322	0.00	1322	0.00	1322
	3.31	MIN	0.00	1322	0.00	1322	0.00	1322	0.00	1322
	4.41	MAX	0.00	1322	0.00	1322	0.00	1322	0.00	1322
	5.51	MIN	0.00	1322	0.00	1322	0.00	1322	0.00	1322

MAX/MIN FORCE VALUES FOR MEMB 9005, AMONGST ALL SECT LOCATIONS									
	FY/	DIST	LD	FZ/	DIST	LD	FX	DIST	LD
MAX.	0.00	0.00	1001	0.00	0.00	1001	19.36 C	0.00	1322
MIN.	0.00	0.00	5.51 1322	0.00	0.00	5.51 1322	19.38 T	5.51	1022

9006	0.00	MAX	0.30	1114	0.00	1322	0.00	1322	0.00	1322
	1.09	MIN	0.19	1322	0.00	1322	0.00	1322	0.00	1322
	2.17	MAX	0.00	1322	0.00	1322	0.00	1322	0.00	1322
	3.26	MIN	0.00	1322	0.00	1322	0.00	1322	0.00	1322
	4.35	MAX	0.00	1322	0.00	1322	0.00	1322	0.00	1322
	5.44	MIN	-0.19	1322	0.00	1322	0.00	1322	0.00	1322

STAAD SPACE

--- PAGE NO. 55

MAX/MIN FORCE VALUES FOR MEMB 9006, AMONGST ALL SECT LOCATIONS									
	FY/	DIST	LD	FZ/	DIST	LD	FX	DIST	LD
MAX.	0.30	0.00	1001	0.00	0.00	1001	0.05 T	0.00	1221
MIN.	-0.30	0.00	5.44	1322	0.00	5.44	1322	8.22 T	5.44 1114

9007	0.00	MAX	0.30	1114	0.00	1322	0.00	1322	0.00	1322
	1.09	MIN	0.19	1322	0.00	1322	0.00	1322	0.00	1322
	2.17	MAX	0.00	1322	0.00	1322	0.00	1322	0.00	1322
	3.26	MIN	0.00	1322	0.00	1322	0.00	1322	0.00	1322
	4.35	MAX	0.00	1322	0.00	1322	0.00	1322	0.00	1322
	5.44	MIN	-0.19	1322	0.00	1322	0.00	1322	0.00	1322

Page 34

[illegible]

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ PX	CRITICAL COND/ MW	RATIO/ MZ	LOADING/ LOCATION
2002	TAPERED	PASS 11.49 C	BS-4.7 (C) -0.12	0.118 -24.46	2011
2003	TAPERED	PASS 4.95 C	BS-4.3.6 -0.17	0.202 44.86	2011
2004	TAPERED	PASS 4.95 C	BS-4.3.6 0.00	0.170 44.86	2013
3001	TAPERED	PASS 0.58 T	BS-4.3.6 0.00	0.131 36.31	2013
3002	TAPERED	PASS 0.58 T	BS-4.3.6 0.00	0.118 36.31	2013
3002	TAPERED	PASS 11.49 C	BS-4.7 (C) -0.12	0.118 -24.46	2011
3003	TAPERED	PASS 4.95 C	BS-4.3.6 0.17	0.202 44.86	2011
3004	TAPERED	PASS 0.58 T	BS-4.3.6 0.00	0.170 26.31	2013
4001	TAPERED	PASS 0.52 T	BS-4.3.6 0.00	0.150 23.18	2013
4002	TAPERED	PASS 4.33 C	BS-4.3.6 0.06	0.158 32.85	2013
4003	TAPERED	PASS 4.33 C	BS-4.3.6 0.06	0.158 32.85	2013
4004	TAPERED	PASS 0.52 T	BS-4.3.6 0.00	0.150 33.18	2013
5001	TAPERED	PASS 0.24 T	BS-4.8.2.2 -11.20	0.069 0.00	2001
5002	TAPERED	PASS 4.40 C	BS-4.7 (C) -15.72	0.101 -0.68	2011
5003	TAPERED	PASS 5.58 C	BS-4.7 (C) -21.36	0.140 -1.99	2011
5004	TAPERED	PASS 0.24 T	BS-4.8.2.2 11.20	0.069 0.00	2001
6001	TAPERED	PASS 0.24 T	BS-4.8.2.2 11.20	0.069 0.00	2001
6002	TAPERED	PASS 4.40 C	BS-4.7 (C) -15.72	0.101 -0.68	2011
6003	TAPERED	PASS 5.58 C	BS-4.7 (C) -21.36	0.140 -1.99	2011
6004	TAPERED	PASS 0.24 T	BS-4.8.2.2 11.20	0.069 0.00	2001
7005 ST	UA90X90X6	PASS 7.03 T	BS-4.6 (T) 0.00	0.019 0.00	2011
7006 ST	UA90X90X6	PASS 5.87 T	BS-4.6 (T) 0.00	0.016 0.00	2014
7007 ST	UA90X90X6	PASS 6.74 T	BS-4.6 (T) 0.00	0.018 0.00	2014
7008 ST	UA90X90X6	PASS 0.03 T	BS-4.6 (T) 0.00	0.000 0.00	2001
STAND SPACE					-- PAGE NO

MEMBER	TABLE	RESULT/ FX	CRITICAL MY	COND/ NY	RATIO/ NZ	LOADING/ LOCATION
8001 ST UA90X90X6	PASS	2.83 T	85-4.6	(T)	0.008	2011
8002 ST UA90X90X6	PASS	0.58 T	85-4.6	(T)	0.002	2014
8003 ST UA90X90X6	PASS	3.17 T	85-4.6	(T)	0.009	2012
				0.00	0.00	0.00

Page 42

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ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
307	TAPERED	PASS 24.68 T	BS-4.8,2.2 0.00	0.040 -12.87	2013 2.00
308	TAPERED	PASS 24.86 T	BS-4.8,2.2 0.00	0.096 34.63	2013 3.13
309	TAPERED	PASS 17.43 T	BS-4.8,2.2 0.00	0.094 -34.60	2013 0.00
310	TAPERED	PASS 1.34 C	BS-4.2,3-(Y) 0.00	0.043 -14.89	2013 0.00
311	TAPERED	PASS 17.80 C	BS-4.7 (C) 0.00	0.051 -17.69	2013 0.82
312	TAPERED	PASS 33.67 C	BS-4.7 (C) 19.44	0.156 -8.64	2011 0.00
401	TAPERED	PASS 52.32 C	BS-4.7,3 (C) 15.27	0.141 13.67	2011 4.04
402	TAPERED	PASS 28.98 C	BS-4.7,3 (C) 15.27	0.154 15.24	2011 0.00
403	TAPERED	PASS 13.26 C	BS-4.2,3 (Y) 0.00	0.042 10.67	2013 0.82
404	TAPERED	PASS 4.46 T	BS-4.8,2.2 -0.88	0.096 34.53	2013 0.82
405	TAPERED	PASS 14.77 T	BS-4.8,2.2 0.74	0.098 -34.53	2013 0.00
406	TAPERED	PASS 14.24 T	BS-4.8,2.2 -0.11	0.033 10.97	2013 0.00
407	TAPERED	PASS 14.24 T	BS-4.8,2.2 0.11	0.033 -10.97	2013 2.00
408	TAPERED	PASS 14.77 T	BS-4.8,2.2 -0.74	0.098 34.53	2013 3.13
409	TAPERED	PASS 4.46 T	BS-4.8,2.2 0.88	0.096 -34.53	2013 0.00
410	TAPERED	PASS 13.26 C	BS-4.2,3 (Y) 0.00	0.042 10.67	2013 0.00
411	TAPERED	PASS 28.98 C	BS-4.7,3 (C) 15.27	0.067 -15.24	2011 0.82
412	TAPERED	PASS 52.32 C	BS-4.7,3 (C) 13.26	0.141 -15.27	2011 0.00

8004	ST	UA90X90X6	PASS	BS-4.6	(T)	0.002	2013
			0.83	T	0.00	0.00	5.90
8005	ST	UA90X90X6	PASS	BS-4.6	(T)	0.002	2013
			0.57	T	0.00	0.00	6.40
8006	ST	UA90X90X6	PASS	BS-4.6	(T)	0.002	2013
			0.57	T	0.00	0.00	6.40
8007	ST	UA90X90X6	PASS	BS-4.6	(T)	0.002	2013
			0.57	T	0.00	0.00	0.00
8008	ST	UA90X90X6	PASS	BS-4.6	(T)	0.002	2013
			0.57	T	0.00	0.00	0.00
8009	ST	UA90X90X6	PASS	BS-4.6	(T)	0.002	2014
			0.58	T	0.00	0.00	0.00
8010	ST	UA90X90X6	PASS	BS-4.6	(T)	0.008	2011
			2.83	T	0.00	0.00	0.00
8011	ST	UA90X90X6	PASS	BS-4.6	(T)	0.002	2013
			0.83	T	0.00	0.00	0.00
8012	ST	UA90X90X6	PASS	BS-4.6	(T)	0.003	2012
			3.17	T	0.00	0.00	0.00
9005	ST	UA90X90X6	PASS	BS-4.6	(T)	0.016	2011
			7.01	T	0.00	0.00	0.00
9006	ST	UA90X90X6	PASS	BS-4.6	(T)	0.016	2014
			5.87	T	0.00	0.00	5.44
9007	ST	UA90X90X6	PASS	BS-4.6	(T)	0.018	2011
			6.74	T	0.00	0.00	5.44
9008	ST	UA90X90X6	PASS	BS-4.6	(T)	0.000	2001
			0.05	T	0.00	0.00	0.00

***** END OF TABULATED RESULT OF DESIGN *****

232. STEEL TAKE OFF ALL
 1 STEEL TAKE OFF J

1	PROFILE	LENGTH(METE)	WEIGHT(KN)
	ISECT: TAPERED	26	135.33
	ISECT: TAPERED	1001	106.038
	ISECT: TAPERED	2002	12.616
	ST UA90X90X6		20.01
			8.730
			116.62
			9.496

		TOTAL =	136.881

***** END OF DATA FROM INTERNAL STORAGE *****

233. FINISH

***** END OF THE STAAD.PRO RUN *****

2 FOUNDATION DESIGN CHECK

Document No.: **DM001-E-AAK-CDP-DR-DCC-373906**

Rev A3

Date: 02 May 07

Page

Calculation/Sketch

Meinhardt Infrastructure Pte Ltd
Consulting Engineers, Planners & Managers



Site	Auxiliary Depot Jebel Ali - Building 21	Date	Apr '07	Job No.	068010
	Footing Summary				
		Designed by	Rudy	Sheet No.	

Soil Bearing Pressure	=	450	kN/m ²
Soil Density	=	20	kN/m ³
Concrete Density	=	24	kN/m ³
Footing Depth below slab soffit	=	1	m
Column Stump Area	=	0.25	m ²

Individual Footing

Column	Column Reaction (ULS)	Soil Load (kN)	Footing Load (kN)	Total SLS Load (kN)	ULS Load (kN)	Footing size			Area of Footing (m ²)	Soil Pressure (kN/m ²)	Factor	Remarks
						L (m)	B (m)	H (m)				
201	69	34.2	28.224	120	156	1.400	1.400	0.600	1.96	61	7.35	OK

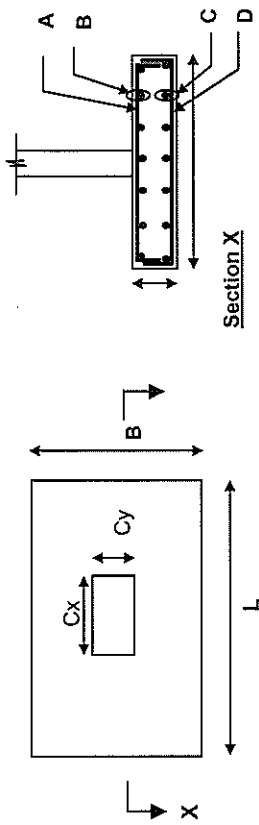
Calculation/Sketch

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Site	Auxiliary Depot Jebel Ali - Building 21 Footing Detail	Date	Apr '07	Job No.	068010
		Designed by	Rudy	Sheet No.	

Summary



Footing Type	L	B	H	Soil Bearing pressure (kPa)	A	B	C	D	Spacing (mm)	Spacing (mm)
F3	1400	1400	600	450	10T12	10T12	10T16	10T16	143	143

Calculation/Sketch

Meinhardt Infrastructure Pte Ltd
Consulting Engineers, Planners & Managers



Site	Auxiliary Depot Jebel Ali - Building 21	Date	Apr '07	Job No.	068010.01
	Footing F3				
	For soil bearing pressure = 150kPa	Designed by	Rudy	Sheet No.	

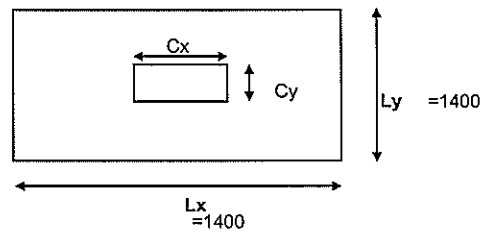
REINFORCED CONCRETE FOOTING DESIGN TO BS8110

1. Material Properties

Concrete grade	f_{cu}	=	35	N/mm ²
Steel grade	f_y	=	460	N/mm ²
Concrete density	γ_c	=	24	KN/m ³
Steel density	γ_s	=	78	KN/m ³

2. Design Data

Depth of Footing	h	=	600	mm
Width of Footing	L_y	=	1400	mm
Length of Footing	L_x	=	1400	mm
Width of Column	C_y	=	450	mm
Length of Column	C_x	=	450	mm
Concrete Ten. Cover	c	=	50	mm
Concrete Comp. Cover	c	=	50	mm
Tension bar dia. (x-dir)	D_x	=	16	mm
Tension bar dia. (y-dir)	D_y	=	16	mm
Comp. bar diameter (x-dir)	D_c	=	12	mm
Comp. bar diameter (y-dir)	D_c	=	12	mm
Effect.depth (x-dir)	d_x	=	542	mm
Effect.depth (y-dir)	d_y	=	526	mm
Area of footing	A	=	1.96	m ²



3. Loadings

Safe bearing pressure	P_b	=	450	KN/m
Load factor	f	=	1.5	
Total axial load (SLS)	N	=	882.00	kN
Eccentricity	e	=	0	mm

4. Analysis

Net pressure (SLS)	P_{max}	=	435.60	KN/m				
Net pressure (ULS)	P_{max}	=	653.4	KN/m				
Shear force (x-dir)	V_x	=	434.51	kN				
Shear force (y-dir)	V_y	=	434.51	kN				
Moment (x-dir)	M_x	=	103.20	kNm				
Moment (y-dir)	M_y	=	103.20	kNm				
Steel required (x-dir)	A_{sreq}	=	1092	mm ²		Nos	ϕ	Spacing
Steel provided (x-dir)	A_{sprov}	=	2011	mm ²	O.K.	10	16	143 x-dir
Steel required (y-dir)	A_{sreq}	=	1092	mm ²		Nos	ϕ	
Steel provided (y-dir)	A_{sprov}	=	2011	mm ²	O.K.	10	16	143 y-dir
Top bars required (x-dir)	A'_{sreq}	=	1092	mm ²		Nos	ϕ	
Top bars provided (x-dir)	A'_{sprov}	=	1131	mm ²	O.K.	10	12	143 x-dir
Top bars required (y-dir)	A'_{sreq}	=	1092	mm ²		Nos	ϕ	
Top bars provided (y-dir)	A'_{sprov}	=	1131	mm ²	O.K.	10	12	143 y-dir

Calculation/Sketch

Meinhardt Infrastructure Pte Ltd
Consulting Engineers, Planners & Managers



Site	Auxiliary Depot Jebel Ali - Building 21	Date	Apr '07	Job No.	068010.01
	Footing F3	Designed by	Rudy	Sheet No.	
	For soil bearing pressure = 150kPa				

5. Shear Check

Checking for punching shear:

Shear force at face $V_f = 1186.31$ kN
Shear stress at face $v_f = 1.22$ N/mm² O.K.

Distance from Support (mm)	av (mm)	V (kN)	v (N/mm ²)	v_e (N/mm ²)	Enhanced v_e (N/mm ²)	Remarks
d	542	0.00	0.00	0.45	0.91	No Shear Links
2d	1084	0.00	0.00	0.45	0.45	No Shear Links

Checking for shear (x-dir):

Shear force at face (x-dir) $V_c = 448.88$ kN
Shear stress at face (x-dir) $v_c = 0.59$ N/mm² O.K.

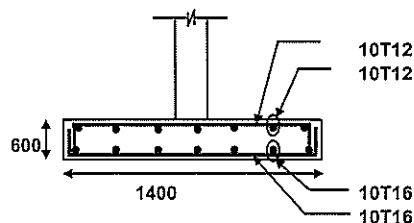
Distance from Support (mm)	av (mm)	V (kN)	v (N/mm ²)	v_e (N/mm ²)	Enhanced v_e (N/mm ²)	Remarks
d	542	0.00	0.00	0.45	0.91	No Shear Links
2d	1084	0.00	0.00	0.45	0.45	No Shear Links

Checking for shear (y-dir):

Shear force at face (y-dir) $V_c = 448.88$ kN
Shear stress at face (y-dir) $v_c = 0.61$ N/mm² O.K.

Distance from Support (mm)	av (mm)	V (kN)	v (N/mm ²)	v_e (N/mm ²)	Enhanced v_e (N/mm ²)	Remarks
d	526	0.00	0.00	0.46	0.92	No Shear Links
2d	1052	0.00	0.00	0.46	0.46	No Shear Links

6. Detailing



3 CONNECTIONS DESIGN

Document No.: **DM001-E-AAK-CDP-DR-DCC-373906**

Rev A3

Date: 02 May 07

Page

Calculation/Sketch

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Site	Dubai Depot Inflammable Material Store (Jebel Ali)	Date	Apr '07	Job No.	068010
Designed by Rudy			Sheet No.		

Column Splice Connection Design

From the STAAD Pro analysis

Fxc =	33	kN (compression)
Fxt =	15	kN (tension)
Fy =	23	kN
Mz =	82	kNm

Total Tension

$$T = (n \cdot T_b / d) \cdot [(d-a) + (d-y_1-a) + (d-y_2-a) + \dots]$$

where n = no of row of bolts and T_b is bolt tension capacity

Total Compression

$$= 0.5a \cdot p_c \cdot b$$

where b = stiff bearing length

$$= \text{web thickness} + 2 \cdot \text{plate thickness}$$

Moment Capacity

$$M_c = [n T_b / (d-a)] \cdot [(d-a)^2 + (d-a-y_1)^2 + (d-a-y_2)^2 + \dots]$$

For Gr 8.8 bolt

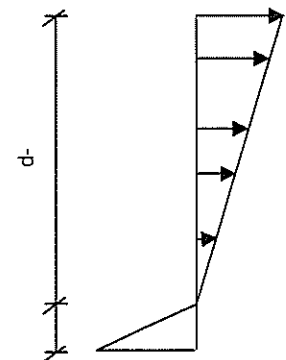
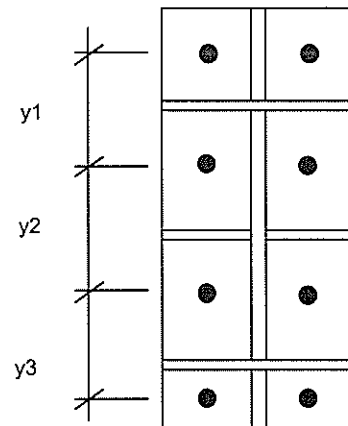
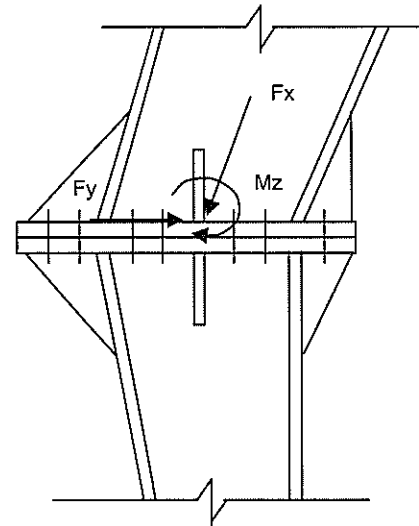
Bolt size	=	20 mm
T _b	=	110 kN
V _b	=	91.9 kN
n	=	2
n total	=	8

web, t _w	=	8 mm
plate, t _p	=	15 mm
b	=	38 mm

y1	127	Trial a	=	64 mm
y2	147	T	=	397.8 kN
y3	127	C	=	397.8 kN
y4	0	T-C	=	0.00 kN
y5	0	M _c	=	105.0 kNm
y6	0			
Depth, d	400			

Tension force on the bolt, T ₁	=	(M _z /M _c) * T _b
	=	86 kN

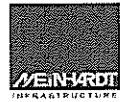
[T ₁ + (F _{xt} /n total)] / T _b + F _y /(n total * V _b) =	0.83
<	1.4 Ok



p_c = 325 N/mm²

Calculation/Sketch

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Consulting Engineers, Planners & Managers



Site	Dubai Depot Inflammable Material Store (Jebel Ali)	Date	Apr '07	Job No.	068010
			Designed by	Rudy	Sheet No.

Check the effective weld length on the stiffener plate to resist bolt's tension

Weld size, s = 6 mm (up to 15mm thick)
Weld strength, pw = 1.05 kN/mm (based on longitudinal capacity)
Weld length required = $(T1 + 0.5 \cdot F_{xt}) / pw$
= 89 mm
= 44 mm per face

Calculation/Sketch

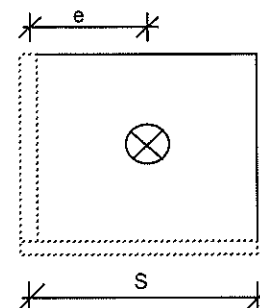
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Site	Dubai Depot Inflammable Material Store (Jebel Ali)	Date	Apr '07	Job No.	068010
			Designed by	Rudy	Sheet No.

From the plate analysis (refer attached) using
Prying Force, $T_o = 10$ kN

Plate width, S_o (mm)	Ave. support bending moment, M_o (kNm)
75	0.020
100	0.027
125	0.035
150	0.040



Actual - Prying force, T	=	86 kN
Side, S	=	75 mm
Bending Moment, M	=	$(T/T_o) * M_o$
	=	0.174 kNm

Bolt Diameter, ϕ	=	20 mm
Bolt distance, e	=	40 mm
Thickness of plate, t_p	=	15 mm
py allow	=	355 N/mm ²

Effective width of the plate resisting the bending moment, b_p	=	$\min (2e + \phi, S)$
	=	75 mm

Section of modulus of the plate, Z_p	=	$b_p * t_p^2 / 6$
	=	2812.5 mm ²

M / Z_p	=	62 N/mm ²	Ok
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Calculation/Sketch

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Site Dubai Depot
Inflammable Material Store (Jebel Ali)

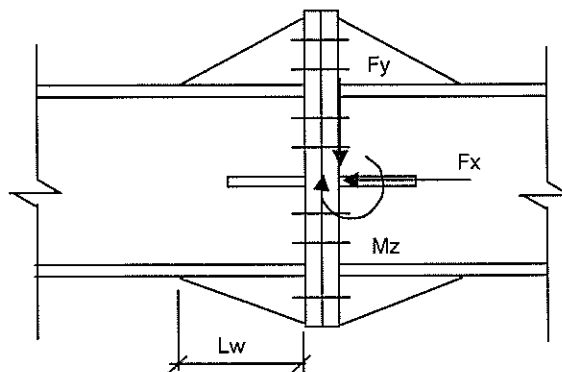
Date Apr '07 Job No. 068010

Designed by Rudy Sheet No.

Beam Splice Connection Design

From the STAAD Pro analysis

F_{xc} = 11 kN (compression)
F_{xt} = 29 kN (tension)
F_y = 31 kN
M_z = 61 kNm



Total Tension

$$T = (n \cdot T_b / d) \cdot [(d-a) + (d-y_1-a) + (d-y_2-a) + \dots]$$

where n = no of row of bolts and T_b is bolt tension capacity

Total Compression

$$= 0.5a \cdot p_c \cdot b$$

where b = stiff bearing length

= web thickness + 2 * plate thickness

Moment Capacity

$$M_c = [n T_b / (d-a)] \cdot [(d-a)^2 + (d-a-y_1)^2 + (d-a-y_2)^2 + \dots]$$

For Gr 8.8 bolt

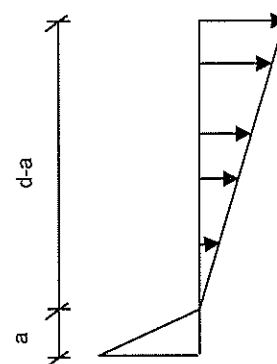
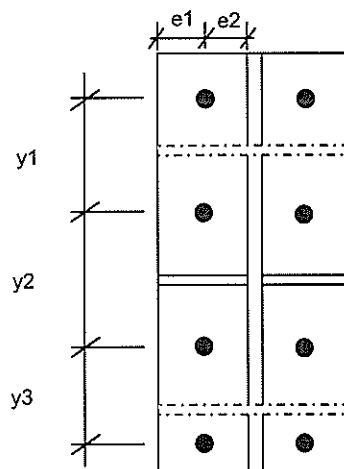
Bolt size = 20 mm
T_b = 110 kN
V_b = 91.9 kN
n = 2
n total = 8

web, t_w = 6 mm
plate, t_p = 15 mm
b = 36 mm
e₁ = 36 mm
e₂ = 36 mm

y ₁	127	Trial a	=	68 mm
y ₂	147	T	=	395.3 kN
y ₃	127	C	=	395.3 kN
y ₄	0	T-C	=	0.00 kN
y ₅	0	M _c	=	103.5 kNm
y ₆	0			
Depth, d	400			

Tension force on the bolt, T₁ = (M_z/M_c) * T_b
= 65 kN

[T₁ + (F_{xt}/n total)] / T_b + F_y/(n total * V_b) = 0.66
< 1.4 Ok



p_c = 325 N/mm²

Calculation/Sketch

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Site	Dubai Depot Inflammable Material Store (Jebel Ali)	Date	Apr '07	Job No.	068010
			Designed by	Rudy	Sheet No.

Check the effective weld length on the stiffener plate to resist bolt's tension

Weld size, s = 6 mm (up to 15mm thick)
Weld strength, pw = 1.05 kN/mm (based on longitudinal capacity)
Weld length required = $(T1 + 0.5 \cdot F_{xt}) / pw$
= 76 mm
= 38 mm per face

Calculation/Sketch

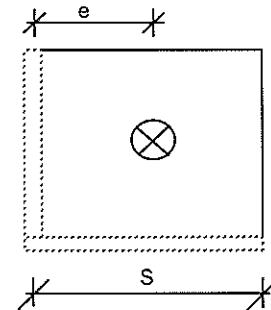
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Site	Dubai Depot Inflammable Material Store (Jebel Ali)	Date	Apr '07	Job No.	068010
		Designed by	Rudy	Sheet No.	

From the plate analysis (refer attached) using
Prying Force, $T_o = 10 \text{ kN}$

Plate width, S_o (mm)	Ave. support bending moment, M_o (kNm)
75	0.020
100	0.027
125	0.035
150	0.040



Actual - Prying force, T	=	68 kN
Side, S	=	100 mm
Bending Moment, M	=	$(T/T_o) * M_o$
	=	0.183 kNm

Bolt Diameter, ϕ	=	20 mm
Bolt distance, e	=	40 mm
Thickness of plate, t_p	=	15 mm
p_y allow	=	355 N/mm^2

Effective width of the plate resisting the bending moment, b_p	=	$\min(2e + \phi, S)$
	=	100 mm

Section of modulus of the plate, Z_p	=	$b_p * t_p^2 / 6$
	=	3750 mm^2

$M / Z_p = 49 \text{ N/mm}^2 \quad \text{Ok}$

Calculation/Sketch

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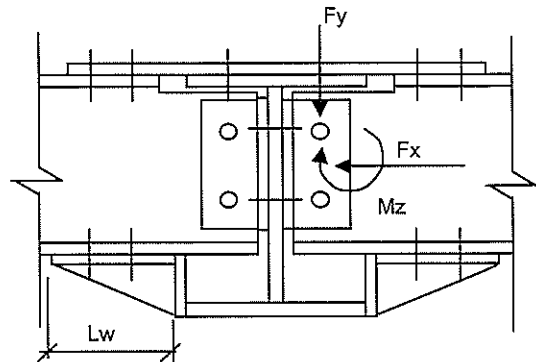
Site	Dubai Depot Inflammable Material Store (Jebel Ali)	Date	Apr '07	Job No.	068010
Designed by Rudy				Sheet No.	

Secondary Beam Connection Design

From the analysis,

$F_y = 35 \text{ kN}$
 $F_x = 17 \text{ kN}$
 $M_z = 71 \text{ kNm}$

Eccentricity of load, $e = 60 \text{ mm}$



For internal span (simply supported span)

- Assume shear due to bending is resisted by top and bottom bolt
- Assume vertical and horizontal shear is resisted by bolt on the web

Check bolts on the flanges

Beam depth - flange thickness, $D_f = 292 \text{ mm}$
 $M_{\max}$ bending moment (incl. moment due to ecc), $M_e = 73.1 \text{ kNm}$

Shear on the flange, $V_f = \frac{M_e}{D_f} = 250.3 \text{ kN}$

No of bolt provided = 4
Shear per bolt = 62.6 kN
Bolt diameter, $\phi = 20 \text{ mm}$
Shear capacity, $V_b = 91.9 \text{ kN}$

Check shear on the web

- for the secondary beam web

No of bolt provided = 2
Shear per bolt = 19.5 kN
Bolt diameter, $\phi = 16 \text{ mm}$
Shear capacity, $V_b = 58.9 \text{ kN}$

- for the main beam web

No of bolt provided = 2
Distance between bolt = 150 mm
Tensile load per bolt = 22.0 kN
Bolt diameter, $\phi = 16 \text{ mm}$
Tensile capacity, $V_b = 70.7 \text{ kN}$

Calculation/Sketch

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Site	Dubai Depot Inflammable Material Store (Jebel Ali)	Date	Apr '07	Job No.	068010
		Designed by	Rudy	Sheet No.	

For the cantilever end

- Assume shear due to bending is resisted by top and bottom bolt
- Assume vertical and horizontal shear is resisted by bolt on the web

Eccentricity of load, e = 60 mm

M_{max} = 32 kNm

V_{max} = 26 kN

Check shear on the web

- for the secondary beam web

No of bolt provided	=	2
Shear per bolt	=	13.0 kN
Bolt diameter, ϕ	=	16 mm
Shear capacity, V_b	=	58.9 kN

- for the main beam web

No of bolt provided	=	2
Distance between bolt	=	150 mm
Tensile load per bolt	=	10.4 kN
Bolt diameter, ϕ	=	16 mm
Tensile capacity, V_b	=	70.7 kN

Check bolts on the top flange

Beam depth - flange thickness, D_f	=	292 mm
M_{max} bending moment (incl. moment due to ecc), M_e	=	33.6 kNm

Shear on the flange, V_f	=	M_e / D_f
	=	114.9 kN

No of bolt provided	=	4
Shear per bolt	=	28.7 kN
Bolt diameter, ϕ	=	16 mm
Shear capacity, V_b	=	58.9 kN

Calculation/Sketch

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Site	Dubai Depot Inflammable Material Store (Jebel Ali)	Date	Apr '07	Job No.	068010
Designed by Rudy				Sheet No.	

Column Splice Connection Design (Supporting the Secondary Beam)

From the STAAD Pro analysis

Fxc =	82	kN (compression)
Fxt =	5	kN (tension)
Fy =	260	kN (include shear due to beam moment)
Mz =	85	kNm

Total Tension

$$T = (n \cdot T_b / d) \cdot [(d-a) + (d-y_1-a) + (d-y_2-a) + \dots]$$

where n = no of row of bolts and Tb is bolt tension capacity

Total Compression

$$= 0.5a \cdot p_c \cdot b$$

where b = stiff bearing length

$$= \text{web thickness} + 2 \cdot \text{plate thickness}$$

Moment Capacity

$$M_c = [n T_b / (d-a)] \cdot [(d-a)^2 + (d-a-y_1)^2 + (d-a-y_2)^2 + \dots]$$

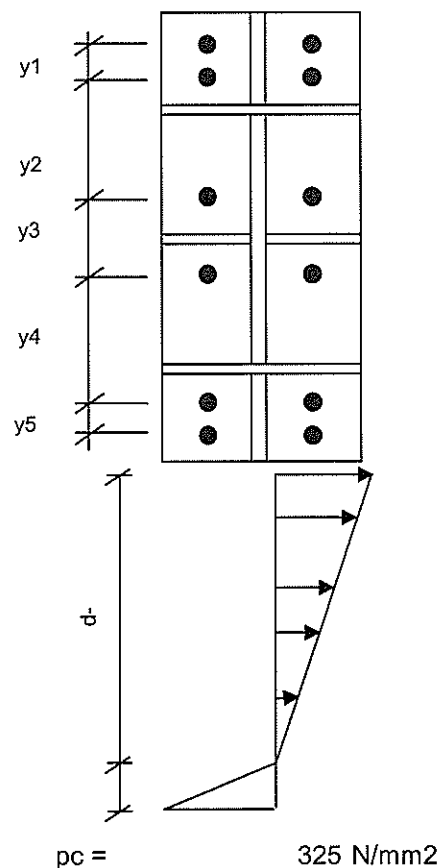
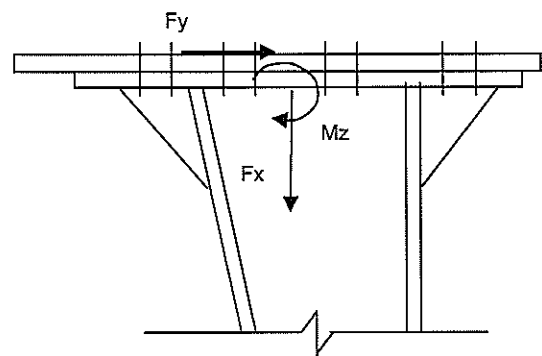
For Gr 8.8 bolt

Bolt size =	20 mm
Tb =	110 kN
Vb =	91.9 kN
n =	2
n total =	8
web, tw =	8 mm
plate, tp =	15 mm
b =	38 mm

y1	50	Trial a	=	94 mm
y2	150	T	=	581.8 kN
y3	100	C	=	581.8 kN
y4	150	T-C	=	0.00 kN
y5	50	Mc	=	186.9 kNm
y6	0			
Depth, d	500			

$$\text{Tension force on the bolt, } T_1 = (M_z / M_c) \cdot T_b = 50 \text{ kN}$$

$$[T_1 + (F_{xt} / n_{\text{total}})] / T_b + F_y / (n_{\text{total}} \cdot V_b) = 0.81 < 1.4 \text{ Ok}$$



Calculation/Sketch

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Site	Dubai Depot Inflammable Material Store (Jebel Ali)	Date	Apr '07	Job No.	068010
			Designed by	Rudy	Sheet No.

Check the effective weld length on the stiffener plate to resist bolt's tension

Weld size, s = 6 mm (up to 15mm thick)
Weld strength, pw = 1.05 kN/mm (based on longitudinal capacity)
Weld length required = $\max(T1 + 0.5 \cdot F_{xt}, F_y) / pw$
= 248 mm
= 124 mm per face

Calculation/Sketch

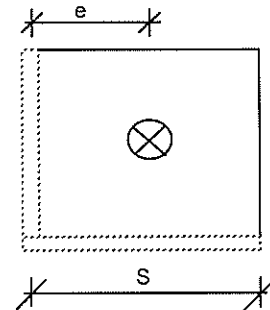
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Site	Dubai Depot Inflammable Material Store (Jebel Ali)	Date	Apr '07	Job No.	068010
Designed by Rudy				Sheet No.	

From the plate analysis (refer attached) using
Prying Force, $T_o = 10$ kN

Plate width, S_o (mm)	Ave. support bending moment, M_o (kNm)
75	0.020
100	0.027
125	0.035
150	0.040



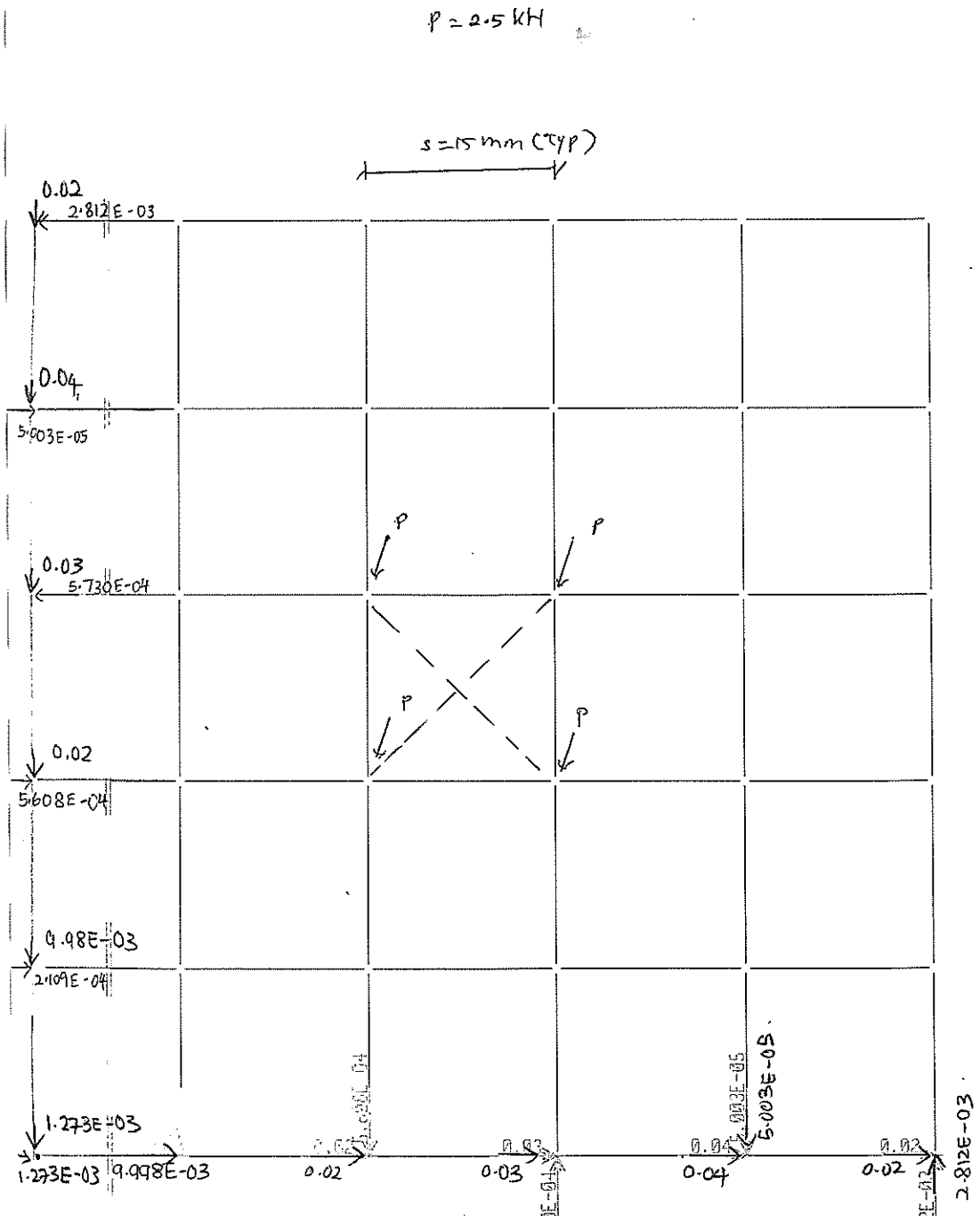
Actual - Prying force, T	=	100 kN
Side, S	=	150 mm
Bending Moment, M	=	$(T/T_o) * M_o$
	=	0.404 kNm

Bolt Diameter, ϕ	=	20 mm
Bolt distance, e	=	30 mm
Thickness of plate, t_p	=	15 mm
p_y allow	=	355 N/mm ²

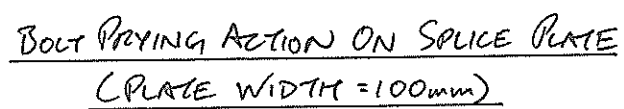
Effective width of the plate resisting the bending moment, b_p	=	$\min(2e + \phi, S)$
	=	80 mm

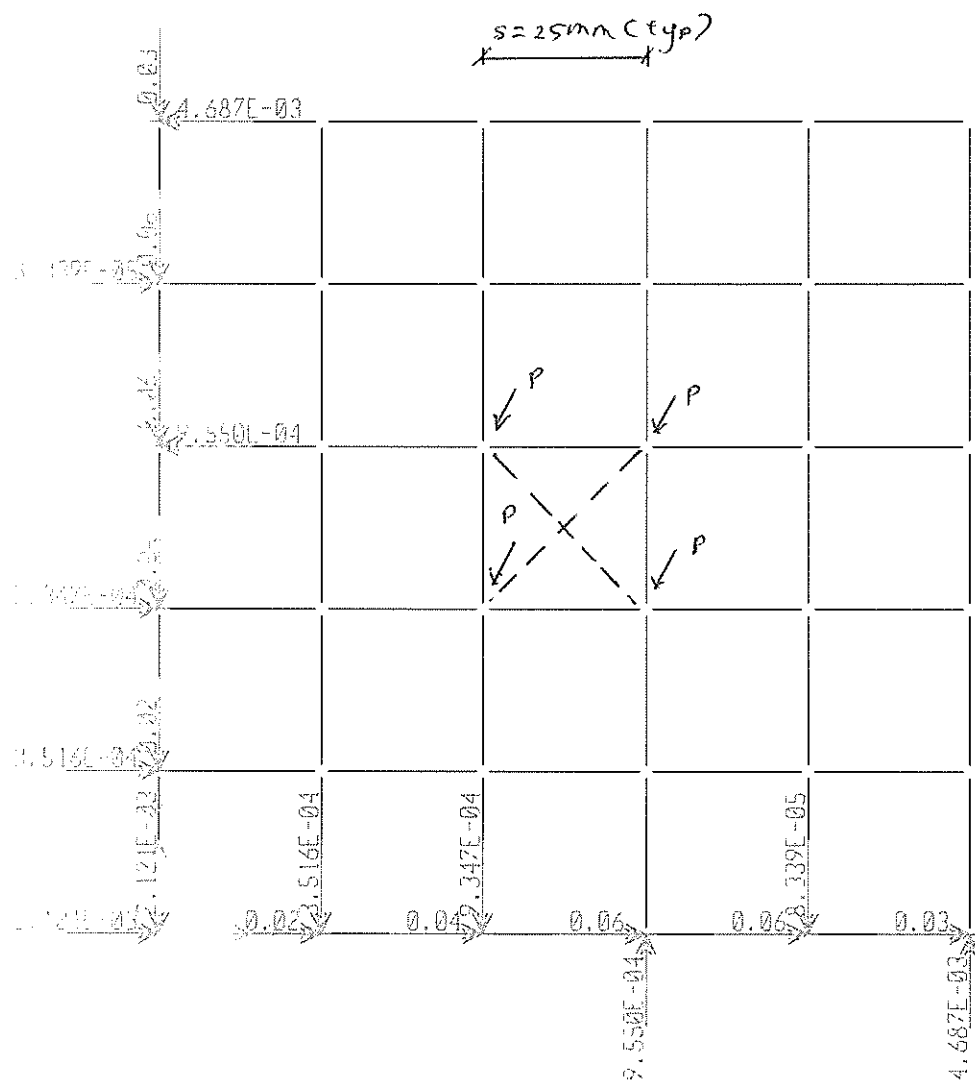
Section of modulus of the plate, Z_p	=	$b_p * t_p^2 / 6$
	=	3000 mm ²

M / Z_p	=	135 N/mm ²	Ok
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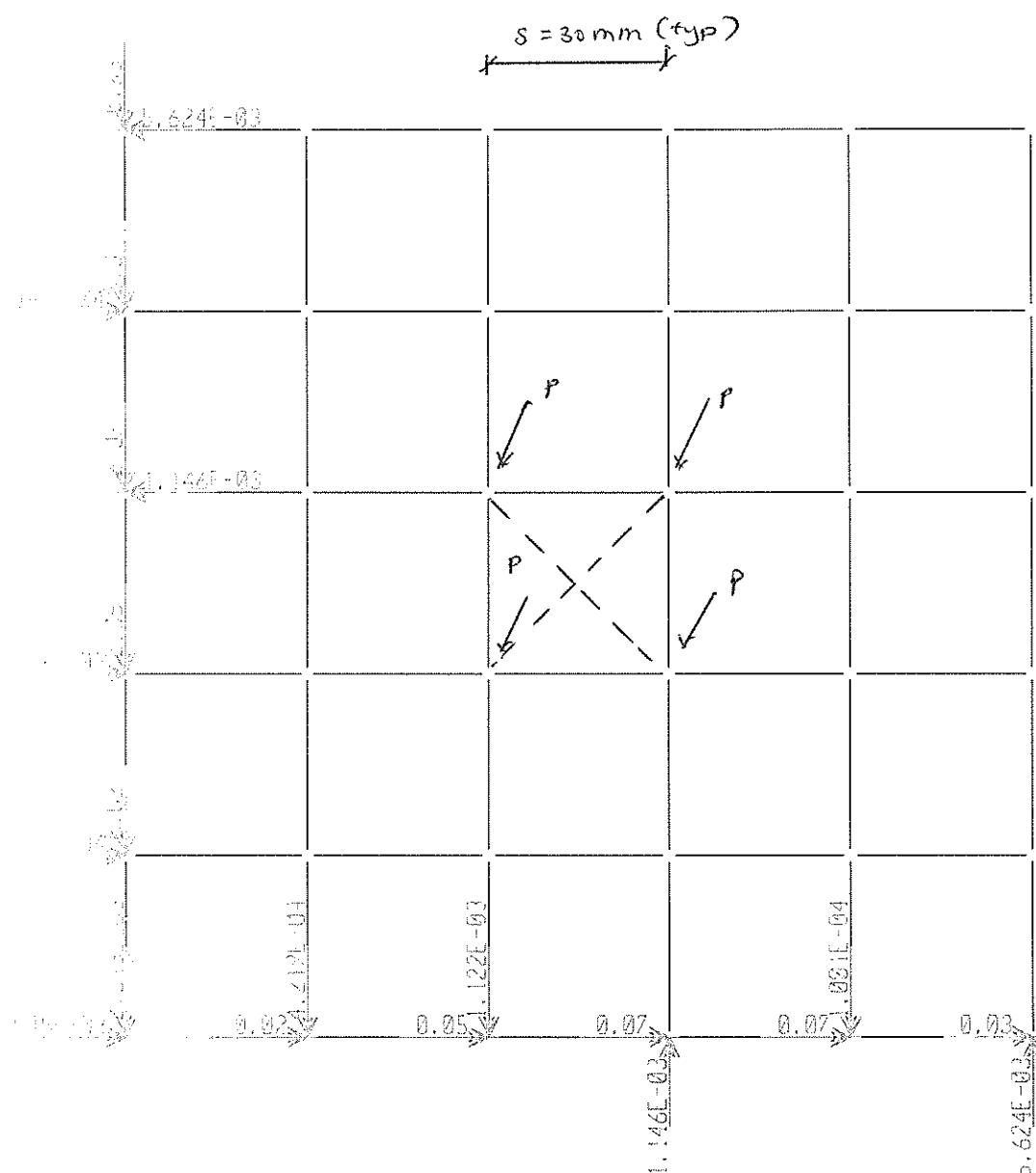
Bolt Prying Action on Splice Plate
(Plate Width = 75 mm)





$$P = 2.5 \text{ kN}$$

BOLT PRYING ACTION ON SPLICE PLATE
(PLATE WIDTH = 125mm)



$$P = 2.5 \text{ kN}$$

BOLT PRYING ACTION ON SPLICE PLATE
(PLATE WIDTH = 150mm)

